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1. Rezumatul Lucrării

Proiectarea și implementarea unei drone de tip cvadricopter cu sistem propriu de control al zborului pentru agricultura de precizie

1.1 Introducere și motivația lucrării

Agricultura reprezintă una dintre cele mai vechi și, totodată, cele mai dinamice ramuri ale activității umane, aflându-se în prezent în fața unor provocări fără precedent. Potrivit estimărilor demografice, populația mondială urmează să atingă aproximativ zece miliarde de locuitori până în anul 2050, ceea ce va impune o creștere a producției alimentare cu aproape 70%. Această cerință se conjugă cu efectele tot mai pronunțate ale schimbărilor climatice, precum seceta, grindina, vânturile puternice și băltirea apei, fenomene care perturbă ciclurile de cultură și diminuează în mod semnificativ productivitatea. În acest context, integrarea tehnologiilor digitale în practica agricolă nu mai constituie o simplă opțiune, ci o necesitate strategică.

Pe fondul acestor presiuni, au apărut numeroase soluții tehnologice menite să sprijine producătorii agricoli. Inteligența artificială generativă prelucrează seturi extinse de date provenite de la senzori, buletine meteorologice și evidențe istorice ale recoltelor, oferind recomandări personalizate; viziunea computerizată permite identificarea în timp real a bolilor, a deficiențelor nutriționale și a buruienilor; conceptul de edge IoT asigură prelucrarea locală a datelor, iar teledetecția prin satelit oferă o monitorizare la scară largă a stării culturilor. La rândul lor, robotica și nanotehnologia contribuie la automatizarea sarcinilor fizice și, respectiv, la eliberarea controlată a substanțelor nutritive. În ansamblul acestor direcții, dronele ocupă un loc distinct, întrucât reunesc mobilitatea, capacitatea de teledetecție și posibilitatea automatizării într-o singură platformă.

Prin definiție, o dronă este un vehicul aerian fără pilot, capabil să opereze autonom sau prin comandă la distanță. În agricultură se utilizează cu precădere trei categorii de drone: cvadricopterele, apreciate pentru manevrabilitatea ridicată; dronele cu aripă fixă, care oferă o autonomie de zbor superioară; și dronele hibride, care combină avantajele celor două tipuri anterioare. Indiferent de configurație, dronele pot fi echipate cu o gamă variată de senzori, de la camere în spectrul vizibil până la camere multispectrale, hiperspectrale, termice și senzori LiDAR.

Lucrarea de față se înscrie în acest domeniu de cercetare și are drept obiectiv proiectarea și construcția, pornind de la nivelul componentelor fundamentale, a unei drone de tip cvadricopter dotate cu un sistem propriu de control al zborului. Spre deosebire de abordările bazate pe controlere comerciale preconfigurate, prezentul demers pune accentul pe dezvoltarea integrală a firmware-

ului, plecând de la principiile sistemelor încorporate de nivel scăzut. Se urmărește astfel nu doar realizarea unei platforme funcționale, ci și dobândirea unei înțelegeri aprofundate a mecanismelor care stau la baza stabilizării zborului și a estimării atitudinii.

1.2 Fundamente teoretice

1.2.1 Arhitectura unei drone agricole

Din perspectivă structurală, o dronă agricolă poate fi descrisă prin intermediul a șase subsisteme distincte, fiecare îndeplinind o funcție specifică. Sistemul de zbor asigură stabilitatea aeronavei și cuprinde structura cadrului, realizată din compozite de fibră de carbon sau din materiale plastice tehnice, precum și sistemul de propulsie, format din motoare de curent continuu fără perii și elicele aferente. Sistemul de sarcină utilă este configurat în funcție de cerințele operaționale și poate include rezervoare pentru pesticide sau semințe, respectiv instrumente de teledetecție destinate monitorizării și cartografierii.

Sistemul de control constituie nucleul inteligenței dronei, fiind responsabil de planificarea traiectoriei, de controlul atitudinii și de execuția sarcinilor; acesta se împarte în sistemul de control al zborului și în stația de sol. Sistemul energetic reunește bateria, de regulă de tip litiu-polimer, și modulul de gestionare a energiei, care distribuie și monitorizează alimentarea subsistemelor. Sistemul de comunicație și de transmisie a datelor mediază schimbul de informații dintre dronă și stația de sol, iar componentele auxiliare, între care sistemele de evitare a obstacolelor, redundanța și disiparea căldurii, contribuie la sporirea siguranței în exploatare.

1.2.2 Vegetația plantelor și principiile imagisticii spectrale

Pentru a înțelege de ce se recurge la camere multispectrale și hiperspectrale în evaluarea stării de sănătate a culturilor, este necesară o scurtă analiză a naturii luminii solare. Radiația solară este compusă din unde de lungimi diferite, dintre care numai o parte este perceptibilă ochiului uman. Plantele răspund însă și la lungimi de undă din afara spectrului vizibil, îndeosebi în domeniile ultraviolet și infraroșu, aspect esențial pentru procesele lor fotochimice.

În cursul fotosintezei, dioxidul de carbon și apa sunt transformate în zaharuri și oxigen prin acțiunea clorofilei, un pigment care absoarbe intens lumina roșie și albastră, dar reflectă lungimile de undă corespunzătoare culorii verzi, motiv pentru care plantele apar verzi. Un aspect deosebit de important îl reprezintă comportamentul plantelor în infraroșul apropiat: o plantă sănătoasă, bogată în clorofilă, reflectă o cantitate mai mare de radiație infraroșie decât una aflată în stare de stres. Analiza spectrului de absorbție și de reflexie oferă, prin urmare, informații valoroase despre

vitalitatea și productivitatea culturilor, informații sintetizate adesea sub forma unor indici de vegetație, precum Indicele de Vegetație prin Diferență Normalizată (NDVI).

Spre deosebire de o cameră convențională, care înregistrează imaginea folosind numai cele trei culori primare, o cameră hiperspectrală descompune lumina incidentă în numeroase benzi spectrale înguste. Rezultatul nu este o imagine unică, ci o structură tridimensională de date, cunoscută sub numele de cub hiperspectral, ale cărei două dimensiuni sunt spațiale, iar cea de-a treia este spectrală. Fiecare pixel al unei asemenea imagini conține un întreg spectru de reflexie, denumit semnătură spectrală; întrucât materiale diferite reflectă și absorb lumina în moduri caracteristice, semnăturile lor spectrale permit deosebirea unor substanțe care ar putea părea identice ochiului uman. În agricultura de precizie, această proprietate face posibilă separarea vegetației sănătoase de cea aflată în stres.

Dispozitivul optic care stă la baza acestei tehnologii este spectrograful imagistic, alcătuit dintr-o fantă de intrare, o optică de colimare, un element dispersiv, de regulă o rețea de difracție, și o lentilă de focalizare. Fanta de intrare limitează lumina la o singură linie a scenei, elementul dispersiv o descompune în spectrul componentelor sale, iar imaginea rezultată este proiectată asupra unui senzor monocrom, care măsoară intensitatea luminii pe fiecare lungime de undă. Imaginea completă a suprafeței vizate este astfel achiziționată linie cu linie. Ansamblul imaginilor captate pe parcursul zborului este ulterior combinat, prin tehnici de fotogrametrie și de aliniere a imaginilor, într-o ortofotografie georeferențiată a întregii parcele, care constituie baza spațială a analizelor ulterioare.

1.3 Implementarea practică

1.3.1 Componentele hardware

Platforma a fost construită în jurul unui cadru DJI F450, o structură larg utilizată în mediul educațional, cu o distanță diagonală între motoare de 450 mm și o masă de aproximativ 280–300 g. Cadrul integrează în placa inferioară o placă de distribuție a energiei, ale cărei trasee de cupru conduc alimentarea direct către regulatoarele electronice de viteză.

Propulsia este asigurată de patru motoare de curent continuu fără perii, de tip 2212 cu o constantă de 920 KV. Echipat cu o elice de 10×4,5 și alimentat de la o baterie 3S, fiecare motor dezvoltă o forță de tracțiune de 800–900 g, astfel încât cele patru motoare generează împreună aproximativ 3,2 kg de tracțiune. Întrucât masa totală a dronei se situează între 1,0 și 1,3 kg, raportul dintre tracțiune și greutate este suficient pentru un zbor stabil. Alimentarea este furnizată de o

baterie litiu-polimer în configurație 3S, cu o tensiune nominală de 11,1 V și o capacitate de descărcare corespunzătoare consumului de vârf al motoarelor.

Fiecare motor este comandat printr-un regulator electronic de viteză de 30 A, care primește semnale PWM de la controlerul de zbor și reglează puterea preluată de la baterie. Legătura radio este realizată printr-un emițător FlySky FS-i6 pe șase canale, funcționând în banda de 2,4 GHz sub protocolul AFHDS 2A, împreună cu receptorul aferent. Elementul central al estimării mișcării este unitatea de măsurare inerțială ICM-20948, un dispozitiv cu nouă axe care integrează, într-un singur cip, un giroscop, un accelerometru și un magnetometru, fiecare pe trei axe.

Alegerea controlerului de zbor a fost precedată de o analiză comparativă a trei platforme de microcontrolere: Arduino Uno, STM32F103C8T6 și STM32F411CEU. Evaluarea a avut în vedere performanța de calcul, capacitatea de memorie, disponibilitatea perifericelor și adecvarea pentru sistemele de control în timp real. Placa STM32F411CEU, denumită „Black Pill”, s-a impus ca soluție optimă, întrucât dispune de un nucleu ARM Cortex-M4 la 100 MHz, de o unitate hardware de calcul în virgulă mobilă și de resurse de memorie extinse, elemente care permit execuția eficientă a buclelor de reglare și a algoritmilor de fuziune senzorială.

1.3.2 Dezvoltarea firmware-ului și estimarea atitudinii

Firmware-ul a fost dezvoltat în mediul STM32CubeIDE, ales pentru accesul direct pe care îl oferă la perifericele hardware și pentru instrumentele avansate de depanare. Dezvoltarea a parcurs mai multe etape succesive: controlul motorului, integrarea unității inerțiale, estimarea atitudinii, reglarea de tip PID și, în final, integrarea comenzii radio.

Controlul motoarelor se bazează pe generarea de semnale PWM prin intermediul întreruperilor de temporizator. Prin configurarea prescalerului și a perioadei temporizatorului au fost obținute o frecvență a întreruperii de 100 Hz și un control precis al factorului de umplere, care determină puterea transmisă motorului. Comunicația cu unitatea inerțială a fost realizată prin protocolul SPI, ale cărui operațiuni de scriere și de citire au fost implementate atât pentru transferuri de un singur octet, cât și pentru transferuri multiple. Pentru a degreva procesorul de sarcina citirii datelor, a fost utilizat mecanismul de acces direct la memorie (DMA), care transferă datele în mod autonom între periferic și memorie, sporind considerabil eficiența și performanța în timp real a sistemului.

Estimarea atitudinii constituie nucleul teoretic al lucrării. Datele accelerometrului și ale magnetometrului sunt mai întâi normalizate, astfel încât să contribuie exclusiv cu informație de orientare, independent de amplitudinea semnalului brut. Din vectorul gravitațional măsurat în

cadrul de referință al corpului pot fi determinate unghiurile de rulu și de tangaj, în timp ce unghiul de rotație, care nu poate fi obținut din accelerometru, este recuperat cu ajutorul magnetometrului. Relația dintre cadrul de referință global, fixat de Pământ, și cadrul mobil, solidar cu aeronava, este exprimată prin intermediul unghiurilor Euler și al matricelor de rotație asociate.

Deoarece unghiurile Euler prezintă o singularitate cunoscută sub numele de blocaj cardanic, care survine atunci când unghiul de tangaj atinge $\pm 90^\circ$, orientarea a fost reprezentată suplimentar prin cuaternioni. Aceștia permit descrierea continuă a oricărei orientări tridimensionale, fără singularități, motiv pentru care constituie soluția standard în estimarea atitudinii pe sisteme încorporate. Cele două categorii de senzori se completează reciproc: accelerometrul și magnetometrul oferă o orientare absolută, dar sensibilă la vibrații și la interferențe, în vreme ce giroscopul furnizează măsurători rapide și nete, afectate însă de o derivă lentă în timp. Combinarea lor printr-un filtru complementar valorifică dinamica rapidă a giroscopului și corectează, prin intermediul accelerometrului și al magnetometrului, deriva pe termen lung.

Controlul propriu-zis al motoarelor este realizat printr-un regulator proporțional-integral-derivativ, în buclă închisă. Spre deosebire de controlul în buclă deschisă, care nu poate compensa perturbațiile, regulatorul PID măsoară continuu ieșirea reală, calculează eroarea față de valoarea de referință și ajustează dinamic semnalul de comandă pentru a anula această eroare. Termenul proporțional determină viteza de răspuns, cel integral elimină eroarea staționară, iar cel derivativ atenuează supracreșterile și oscilațiile. În sfârșit, comanda radio este preluată prin protocolul digital IBUS, care transmite valorile tuturor canalelor pe un singur fir UART, fiecare pachet fiind verificat printr-o sumă de control ce permite detectarea și eliminarea datelor corupte.

1.4 Rezultate experimentale

Sistemul a fost mai întâi asamblat pe o placă de test, unitatea inerțială ICM-20948 fiind conectată la microcontrolerul STM32 prin interfața SPI1, cu linii dedicate pentru selecția cipului, ceasul de sincronizare și transferul bidirecțional al datelor. Achiziția datelor brute de la senzori, cu unitatea inerțială așezată orizontal și în repaus, a evidențiat mai multe abateri caracteristice. Accelerometrul prezenta o eroare sistematică de decalaj, precum și o deviație a factorului de scară, ambele susceptibile să introducă erori în unghiurile de rulu și de tangaj. Giroscopul indica valori apropiate de zero, ceea ce corespunde comportamentului așteptat al unui senzor static. Magnetometrul, în schimb, prezenta decalaje importante pe toate cele trei axe, atribuite distorsiunilor de tip „hard-iron” generate de materialele feromagnetice din vecinătatea senzorului, distorsiuni care ar introduce o eroare sistematică în estimarea unghiului de rotație.

Pentru validarea funcționării algoritmului de estimare a atitudinii a fost dezvoltat un instrument de vizualizare tridimensională, realizat în limbajul Python. Aplicația stabilește o legătură serială cu microcontrolerul prin periferia USART2 și primește unghiurile Euler calculate de filtrul complementar cu o frecvență de 100 Hz, redând în timp real orientarea estimată a dronei sub forma unui poliedru tridimensional. Pe parcursul etapelor de dezvoltare și de integrare, acest instrument a funcționat ca principal mijloc de diagnosticare, permițând verificarea alinierii corecte a axelor de măsurare, a remapării cadrelor de referință, a stabilității numerice a filtrului complementar și a integrității transmisiei de date.

Într-o etapă ulterioară, circuitul a fost transferat de pe placa de test pe un ansamblu lipit permanent, mai robust din punct de vedere mecanic și pregătit pentru zbor. Consolidarea întregii electronici pe o singură placă compactă a redus complexitatea cablajului și a îmbunătățit fiabilitatea sub acțiunea vibrațiilor. În paralel, legătura radio a fost testată în prealabil în mediul Arduino, verificându-se recepția și decodarea corectă a semnalelor emițătorului: fiecare canal a produs o lățime de impuls cuprinsă în intervalul așteptat, de aproximativ 1000–2000 μ s, cu o poziție neutră în jurul valorii de 1500 μ s, ceea ce a confirmat funcționarea corectă a receptorului și consecvența cartografierii canalelor.

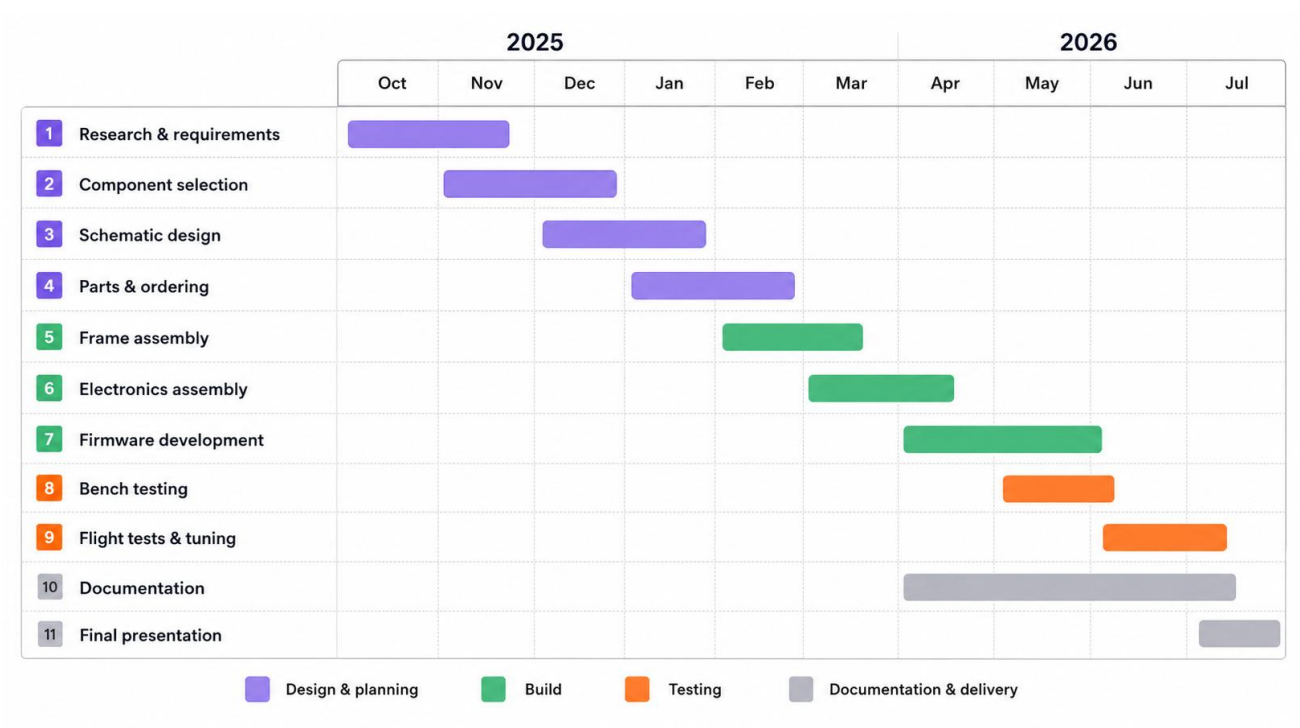
1.5 Concluzii și direcții de dezvoltare

Lucrarea a demonstrat proiectarea și implementarea reușită a unei platforme de tip cvadricopter, construită în jurul microcontrolerului STM32F411CEU și a unității inerțiale ICM-20948, cu un firmware de control al zborului dezvoltat integral pe baza principiilor sistemelor încorporate de nivel scăzut. În cursul acestui demers au fost consolidate competențe în domeniul comunicației SPI, al accesului direct la memorie, al estimării atitudinii în timp real prin filtrare complementară și algebră a cuaternionilor, precum și al reglării PID în buclă închisă. Fundamentele teoretice prezentate oferă, la rândul lor, contextul necesar înțelegerii importanței tehnologiei dronelor în agricultura de precizie.

Implementarea actuală prezintă însă câteva limitări. Absența unui altimetru barometric face ca menținerea altitudinii să depindă exclusiv de comanda manuală, iar lipsa unui modul GPS împiedică navigația autonomă și menținerea poziției. De asemenea, filtrul complementar, deși eficient din punct de vedere computațional, rămâne sensibil la accelerațiile liniare prelungite și la interferențele magnetice, iar autonomia de zbor este limitată de capacitatea bateriei. Nu în ultimul rând, sistemul nu dispune încă de un mecanism de evitare a obstacolelor, ceea ce restrânge exploatarea sigură la spații deschise și controlate.

Sunt prevăzute mai multe direcții de dezvoltare menite să extindă capacitățile platformei. Pe termen scurt, integrarea unui senzor barometric va permite menținerea automată a altitudinii, iar adăugarea unui modul GPS va face posibilă implementarea funcției de întoarcere la punctul de decolare, un mecanism esențial de siguranță. Aceste completări vor marca tranziția de la o aeronavă pilotată manual la una semiautonomă. Pe termen lung, cea mai importantă extindere avută în vedere constă în integrarea unui calculator monoplacă Raspberry Pi, împreună cu un modul de cameră, care va prelua sarcinile de prelucrare de nivel superior. Pe această bază vor fi dezvoltați un algoritm de cartografiere a terenului, capabil să genereze hărți aeriene georeferențiate, și un flux de analiză a stării de sănătate a culturilor, prin calcularea indicilor de vegetație. Împreună, aceste funcționalități vor apropia sistemul de clasa dronelor destinate agriculturii de precizie și vor oferi o platformă solidă pentru cercetări viitoare privind vehiculele aeriene agricole cu cost redus și cu arhitectură deschisă.

Work Planning



2. State of the Art

2.1 History of Agriculture

Over the past two centuries, agriculture, like many other industries, has undergone a profound transformation. Its origins can be traced back to approximately 10,000 BCE, when farming first emerged in the region extending from the eastern Mediterranean coast to the Persian Gulf. Between 4000 and 3000 BCE, irrigation systems were developed in Egypt and Mesopotamia in order to regulate river flows and to secure a reliable water supply for crops. Further advances were achieved during the Bronze Age (3000–1000 BCE), when improvements in metalworking permitted more efficient soil tillage and greater productivity. The British Agricultural Revolution of the eighteenth and nineteenth centuries introduced novel crop-rotation methods and brought previously uncultivated land into production. During the nineteenth and twentieth centuries, the advent of agricultural machinery such as tractors and mechanical harvesters increased efficiency and output dramatically. From the twentieth century to the present day, a broad range of methods and technologies has emerged, including sustainable agriculture, which prioritises environmentally sound and socially responsible practices, and agricultural biotechnology, which employs genetic engineering to enhance crop yields and improve plant traits.

The agricultural sector today faces major challenges that will shape both its near and its more distant future. The foremost of these is meeting the growing demand for larger quantities of higher-quality food. According to a range of estimates, the global population is expected to reach approximately ten billion by 2050, which would require farmers to produce roughly 70% more food¹. At the same time, consumers increasingly demand healthier and more sustainable produce. Farmers must likewise adapt to more frequent extreme weather events and to the changing growing conditions brought about by climate change. Unpredictable rainfall patterns and rising temperatures disrupt conventional growing seasons and diminish agricultural productivity. The principal weather-related difficulties confronting farmers today include the following:

- drought, which depletes soil-moisture reserves and thereby lowers both yield and crop quality;
- hail, which can bruise fruit, tear foliage and break branches, thereby leaving plants vulnerable to pests and disease;
- strong winds, which cause direct damage by stripping leaves and fruit and indirect damage by delaying essential fieldwork;
- waterlogging, which degrades the soil, impairs nutrient uptake, compromises root health and interferes with essential farming operations.

¹ World Economic Forum, 2024 : 5

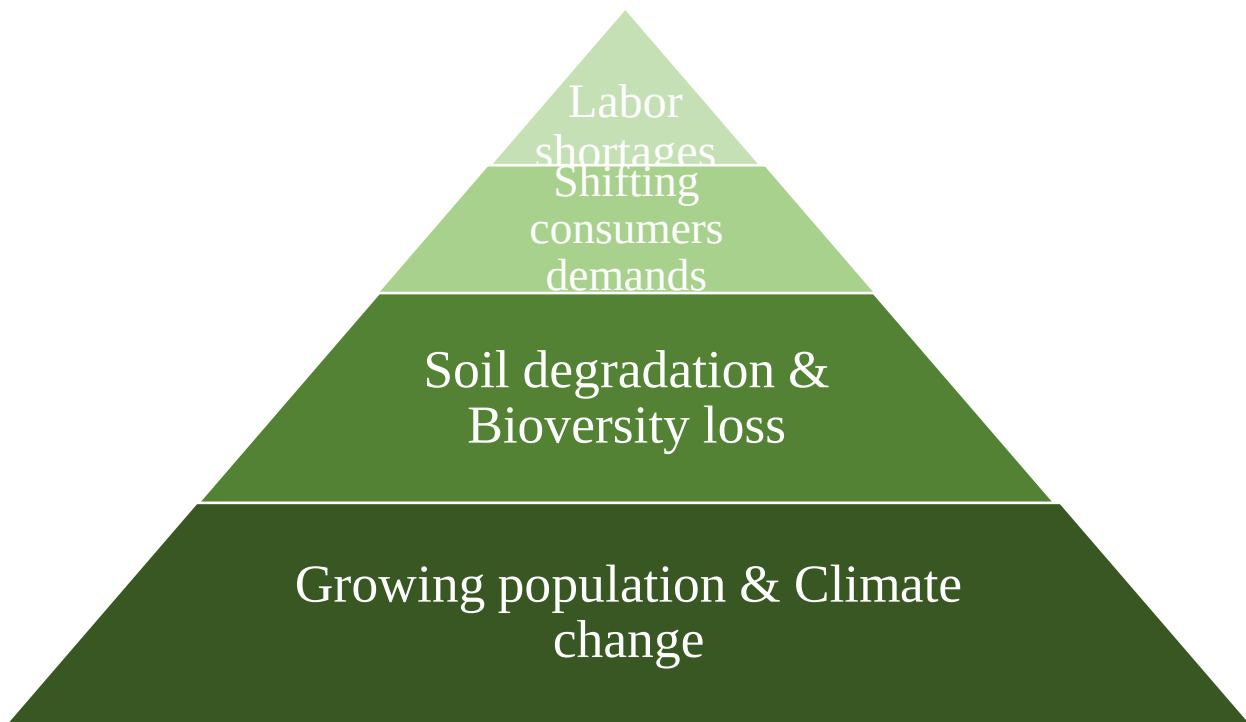


Figure 1 Challenges in agriculture

A number of emerging technologies now assist farmers in producing greater quantities of healthier food:

Generative AI (GenAI) is notable for its capacity to generate new content or data on the basis of existing patterns. Within agritech, it processes large datasets derived from sensors, weather reports and historical yields in order to provide tailored recommendations. Farmers may, for instance, consult large-language-model chatbots to obtain customised guidance on crop rotation or pest control, adapted to local soil and climatic conditions.

Computer vision interprets images and video in order to extract visual information. By training algorithms on labelled crop-image datasets, it enables the real-time identification of plant diseases, nutrient deficiencies, weeds and stages of ripeness. Cameras mounted on tractors or drones scan the fields, supporting automated weeding or harvesting and reducing manual labour, while after harvest the produce can be sorted by quality so as to minimise waste. Its principal advantages are speed and consistency; however, the variability of field conditions, such as changing illumination and differing growth stages, demands robust datasets. High equipment costs and poor connectivity in remote areas remain significant barriers, although edge computing, whereby data are processed locally, helps to overcome some of these limitations.

Edge IoT extends the conventional Internet of Things by performing data analysis directly on the device, which permits rapid responses even in the absence of a cloud connection. Sensors embedded in the soil or in machinery monitor moisture, temperature and pest activity, and on-device algorithms trigger actions such as the automatic opening of irrigation valves. This capability is particularly valuable in rural areas with unreliable internet access, since it enables autonomous farm management. Edge networks can, for example, optimise fertiliser application by responding to real-time conditions and thereby reduce waste.

Satellite-enabled remote sensing provides large-scale monitoring by means of orbital imagery that captures spectral data on crop health and land use. Machine learning improves the prediction of yields and the early detection of water stress. Farmers are supplied with high-resolution maps that allow pest outbreaks to be identified rapidly, while insurers draw on the same data for automated, geo-tagged claim verification. The approach combines frequent satellite passes

with drone flights for finer detail, reducing the need for manual field surveys and increasing supply-chain transparency.

Robotics, encompassing both drones and ground-based machines, automates physical tasks through embedded sensors and artificial intelligence. Drones monitor crops from above, whereas robots equipped with computer vision sow seeds with precision or remove weeds without recourse to chemicals. Swarm robotics allows multiple units to operate cooperatively, sharing data and adapting to changes in terrain. Such solutions help to mitigate labour shortages on large farms and to reduce the soil compaction caused by heavy machinery, while perception algorithms manage obstacle avoidance and decision-making. High initial costs, together with difficulties in handling delicate crops or uneven ground, continue to slow adoption in developing regions, although improved cloud integration is steadily enhancing performance.

Finally, nanotechnology operates at the atomic scale to create precise delivery systems. Nano-fertilisers encapsulate nutrients so as to achieve a controlled release that is synchronised with the plant's growth stages and that reduces runoff, while nano-sensors detect soil contaminants at an early stage. These innovations improve resource efficiency and help to protect the environment. Despite their promise for sustainable farming, uncertainty regarding long-term health effects and a lack of adequate regulation currently limit their widespread adoption, and ongoing research continues to assess their safety and efficacy².

2.2 Drones

A drone, by definition, is an unmanned aerial vehicle (UAV): a broad term denoting an aircraft that operates autonomously or under remote control, without a pilot on board. Drones are typically of small or medium size, and these 'flying robots' are capable of performing a wide range of tasks, from covert military operations to package delivery and aerial photography³.

Beyond military applications, drones are employed across numerous other sectors. In the creative industries, for instance, photographers use them to capture aerial images and video of events and natural scenery. In logistics and transportation they convey goods between locations, and in the aftermath of disasters they support search-and-rescue operations: rather than exposing personnel to hazardous conditions, authorities can deploy autonomous vehicles, including underwater craft, to assist. A further application is wildlife monitoring, in which drones help to track animal populations and their movement patterns. When fitted with LiDAR sensors, drones are also used for three-dimensional modelling, surveying landscapes to produce accurate 3D representations of the terrain. Finally, drones are increasingly applied in agriculture, where they allow farmers to estimate crop yields and monitor livestock, thereby maximising operational efficiency.

Three principal types of drone are employed in agriculture:

Quadcopters, also known as quadrotors, are equipped with four rotors arranged in a square configuration. This arrangement renders the aircraft highly manoeuvrable and enables flight at low altitudes, from which the drone can capture high-resolution imagery and video for monitoring crop health, detecting stress and identifying diseases or pests. Combined with advanced sensors and GPS technology, quadcopters can also be used to map agricultural fields. A further advantage lies in their ability to apply pesticides with precision, which reduces the associated environmental impact. Their limitations, however, include a relatively short flight time (typically 20–30 minutes per flight) and a restricted payload capacity.

² World Economic Forum, 2025 : 4

³ <https://builtin.com/drones>

A second category comprises fixed-wing drones, whose configuration resembles that of a conventional aeroplane and which generate lift through forward motion. This design affords considerably greater flight endurance. Such drones are employed for aerial mapping and crop scouting and are able to cover extensive areas, since they can remain airborne for up to two hours. Their principal drawback is that they require a runway for take-off and landing, which is often impractical in densely populated areas, and they are less manoeuvrable than quadcopters.

The third category consists of hybrid drones, which combine the characteristics of quadcopters and fixed-wing aircraft. Their vertical take-off and landing (VTOL) capability allows them to cover large distances, and they are widely used for crop monitoring, mapping and precision agriculture, proving well suited to farming environments of every scale. Their chief advantage is the flexibility of their flight modes, permitting both low-altitude hovering and the coverage of extensive areas. Their principal limitation is that, owing to their advanced control systems, they are both expensive and highly complex⁴.

In agricultural applications, a variety of sensors and cameras may be mounted on drones in order to monitor crop parameters such as temperature, moisture content and canopy height.

Red-Green-Blue Cameras: These cameras are widely used on account of their low cost and their ability to capture high-resolution imagery⁵. They afford a clear visual overview of the farmland, supporting the assessment of crop growth, the estimation of biomass and the monitoring of visually detectable problems, and they operate effectively under a wide range of weather conditions.

Multispectral and Hyperspectral Cameras: These sensors detect light beyond the visible spectrum, notably in the near-infrared, in order to characterise the reflectance patterns associated with crop health. From these measurements they compute the Normalised Difference Vegetation Index (NDVI) and other vegetation indices, enabling farmers to map areas in which crops are flourishing and those in which they are under stress⁶. The distinction between the two lies in their spectral resolution: a multispectral camera records between five and twelve spectral bands, whereas a hyperspectral camera captures hundreds or even thousands of much narrower bands.

Thermal Sensors: These sensors measure temperature differences across the field in order to reveal thermal patterns in the soil and vegetation. By identifying zones that are excessively dry or over-watered, they help farmers to devise more efficient irrigation strategies.

LiDAR (Light Detection and Ranging) Sensor: This sensor emits laser pulses to measure the geometry of the terrain and to generate accurate three-dimensional maps. It supplies topographical information on elevation, slope and surface irregularities, all of which influence the way water, fertilisers and sprays are distributed across a field.

Spectroradiometer: This instrument quantifies light reflectance across a range of wavelengths, collecting spectral reflectance data from vegetation in particular. It combines the principles of spectroscopy, which examines the interaction between light and matter, with those of radiometry, which is concerned with the measurement of electromagnetic radiation. By integrating these principles, the device evaluates the light reflected from vegetation, a measurement that is essential for the calculation of vegetation indices⁷.

⁴ https://terra-droneagri.com/exploring-the-3-common-types-of-drones-used-in-agriculture/?srltid=AfmBOor3pA-A9nEDYhRWRxH8kAOs3NL1r8lg89UzR96RMV_Jm6vc9vjA

⁵ <https://onlinelibrary.wiley.com/doi/full/10.1002/moda.70021>

⁶ <https://www.leher.ag/blog/drone-sensors-modern-agriculture>

⁷ <https://onlinelibrary.wiley.com/doi/full/10.1002/moda.70021>

3. Theoretical Fundamentals



Figure 1: Components of a drone

An agricultural drone may be described in terms of six distinct subsystems, each fulfilling a specific function: the flight system, the mission-payload system, the control system, the energy system, the communication and data-transmission system, and the auxiliary components.

The purpose of the flight system is to ensure stable flight, which constitutes the foundation of all subsequent operation. It comprises two principal elements: the frame structure and the power system. The frame is fabricated either from carbon-fibre composites, which are valued for their low mass and high strength, or from engineering plastics, whose principal advantage is their low cost. Its configuration may be multi-rotor (with four or six rotors), fixed-wing or hybrid, and it incorporates dedicated mounting points for the mission payload. The power system contains the motors, which are predominantly brushless DC motors on account of their high efficiency and long service life; the number of motors corresponds to the number of propellers, and motors rated at 200–300 W are typically employed, depending on the load requirements. The power system further includes an Electronic Speed Controller (ESC), which links each motor to the flight-control system and regulates motor speed in accordance with flight control commands, controls rotor lift, and enables attitude adjustments⁸. The power system also includes the propellers, which are manufactured from carbon fibre or nylon.

The mission-payload system is configured according to operational requirements and comprises two categories: plant-protection payloads and monitoring or mapping payloads. Plant-protection payloads may include tanks for storing pesticides, seeds or fertilisers, together with a spraying system governed by the flight controller or a sowing mechanism for seeds and fertilisers. Monitoring and mapping payloads, by contrast, may comprise remote-sensing instruments

⁸ <https://www.lintechtt.com/components-of-agricultural-drones>

(multispectral, hyperspectral and thermal-imaging cameras), GPS/RTK modules for precision mapping, and LiDAR.

The control system is responsible for path planning, attitude control and task execution, and thus constitutes the core of the drone's intelligence. It is divided into the flight-control system and the ground-station system. The flight-control system is built around a core module integrating a Microcontroller Unit (MCU) or System on Chip (SoC), whose function is to process sensor data and to issue control commands. It incorporates a number of integrated sensors, among them an Inertial Measurement Unit (IMU) that enables stable hovering. This unit contains accelerometers, gyroscopes and magnetometers, which together monitor the drone's attitude in real time. The gyroscope measures the rate of rotation and thereby helps to keep the drone balanced. In essence, a gyroscope consists of a spinning wheel mounted on an axis that is free to move in any direction, and it is used to provide stability or to maintain a reference direction⁹.

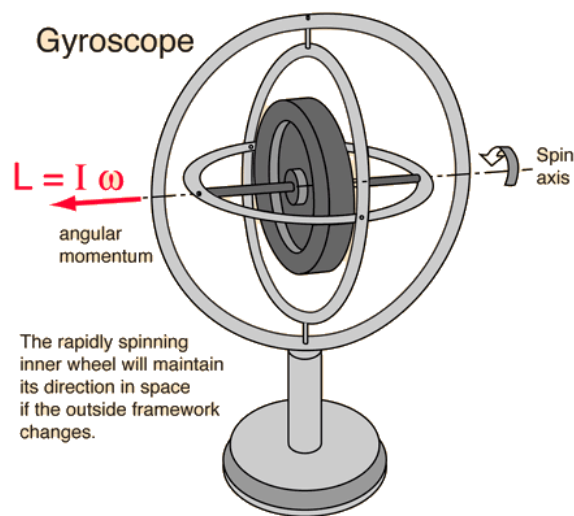


Figure 2 Gyroscope

The accelerometer measures the rate of change of an object's velocity over time¹⁰. Magnetometers, in turn, are devices used to measure magnetic fields¹¹.

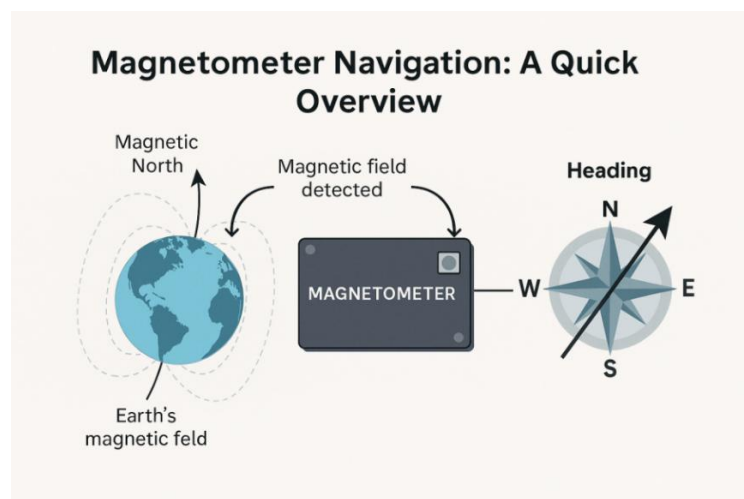


Figure 3 Magnetometer Navigation: A quick overview

⁹ <https://www.jameco.com/Jameco/workshop/ProductNews/howitworks-drone.html?srsId=AfmBOoqxCPoJ2GsgB8RnuLsUnhZ8hgToD1HFXyAgMkndU-QCDQYAwgW>

¹⁰ <https://www.epsondevice.com/sensing/en/tech/column/accelerometer/>

¹¹ <https://www.pnisor.com/understanding-magnetometers-and-their-uses/>

The system also incorporates a barometer/altimeter for the regulation of flight altitude, for example maintaining a height of 1.5–3 metres above the ground during spraying operations.

It further contains GPS/BeiDou modules, which provide positioning and navigation and thereby support autonomous route planning. BeiDou is a Chinese satellite positioning system whose global constellation allows users to obtain navigation and timing data at any location on Earth¹². The ground-station system encompasses both hardware and software functions. Its hardware typically takes the form of mobile applications, tablets or dedicated remote controllers, which communicate with the drone over wireless links such as 4G, Bluetooth or dedicated data-transmission radios. Its software supports map annotation (for instance the delimitation of operating areas), route planning (such as the setting of flight speed, altitude and spraying rate), real-time monitoring (for example of battery level and the remaining volume in the pesticide tank), and the logging of operations (including statistics on treated area and usage).

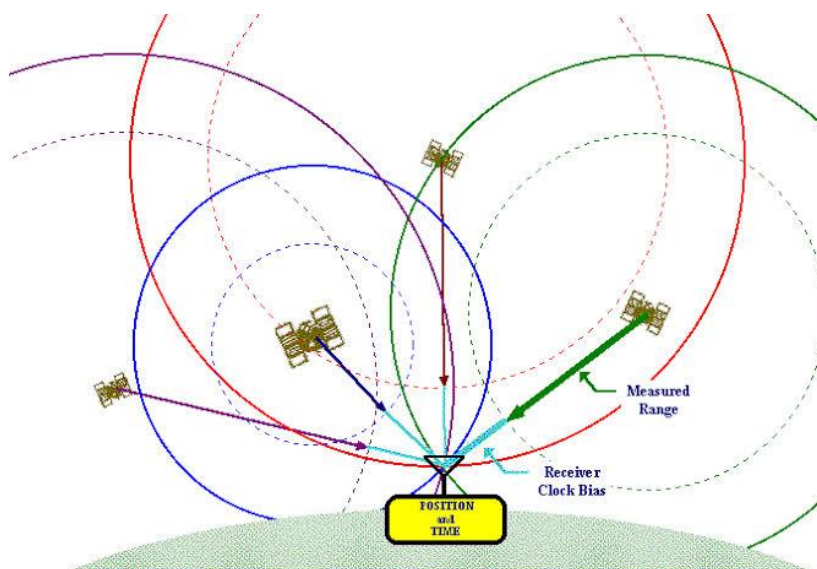


Figure 4 GPS module

The energy system consists of two core components: the battery and the power-management unit (PMU), which together balance endurance against load capacity. In agriculture, the batteries most frequently used are lithium-polymer (LiPo) cells, whose high energy density (of the order of 200–250 Wh/kg) and large discharge current are well suited to powering high-power motors. Their specifications are chosen according to the load requirements: ten-kilogram-class drones, for example, commonly employ 6S lithium batteries (six cells in series) with a capacity of 10,000–20,000 mAh and an endurance of 15–25 minutes, whereas larger, thirty-kilogram-class models may connect several batteries in parallel to achieve an endurance of 30–40 minutes. The protection circuitry typically guards against overcharge, over-discharge and short circuit, and some designs additionally provide thermal management to prevent high-temperature swelling¹³. The power-management unit distributes electrical power from the battery to the principal subsystems, including the flight controller, the propulsion units and the mission payload. It continuously monitors the voltage and current drawn by each subsystem so as to maintain a stable and reliable supply. Should the battery level fall below a predefined threshold, typically around 20%, the unit automatically issues a return command to ensure the safe recovery of the drone.

The communication and data-transmission system governs the exchange of information between the drone and the ground station and provides the operator with continuous control of the

¹² <https://www.sbg-systems.com/glossary/beidou-navigation-satellite/>

¹³ <https://www.lintechtt.com/components-of-agricultural-drones/>

aircraft. It consists of two principal parts: the data-transmission modules and the image-transmission modules. The data-transmission modules convey control commands, such as adjustments to altitude and speed, as well as status information, including position and battery level; they generally operate in the 2.4 GHz or 900 MHz bands and offer a range of approximately one to five kilometres. The image-transmission modules relay real-time imagery from the drone's camera, allowing the operator to detect untreated areas or obstacles within the working zone; these commonly operate in the 5.8 GHz band with a latency of less than 100 ms.

A number of auxiliary elements are provided in order to enhance safety, and these fall into three groups: the obstacle-avoidance system, the redundancy design and the heat-dissipation system. The obstacle-avoidance system employs a range of sensors to detect obstacles and prevent collisions, most commonly millimetre-wave radar (for example at 24 GHz), ultrasonic sensors and vision cameras. The redundancy design provides critical components with duplicate backups, so that if one component fails the system automatically switches to its backup, thereby improving reliability. The heat-dissipation system serves heat-generating components such as the motors, ESCs and flight controller, employing heat sinks, fans or air ducts to prevent elevated temperatures from degrading performance, particularly during extended summer operations.

3.1 Vegetation of the Plants

In order to appreciate why farmers employ multispectral and hyperspectral cameras to assess plant health, it is first necessary to consider the nature of sunlight itself.

Over billions of years, sunlight has continuously acted upon the Earth. Sunlight is, by definition, the electromagnetic radiation, or energy, emitted by the Sun. It is composed of waves of differing wavelength, a subset of which corresponds to the colours perceptible to the human eye. The Sun emits radiation across almost the entire spectrum, including wavelengths that the human eye cannot detect, such as radio waves, microwaves, infrared (IR), ultraviolet (UV), X-rays and gamma rays¹⁴.

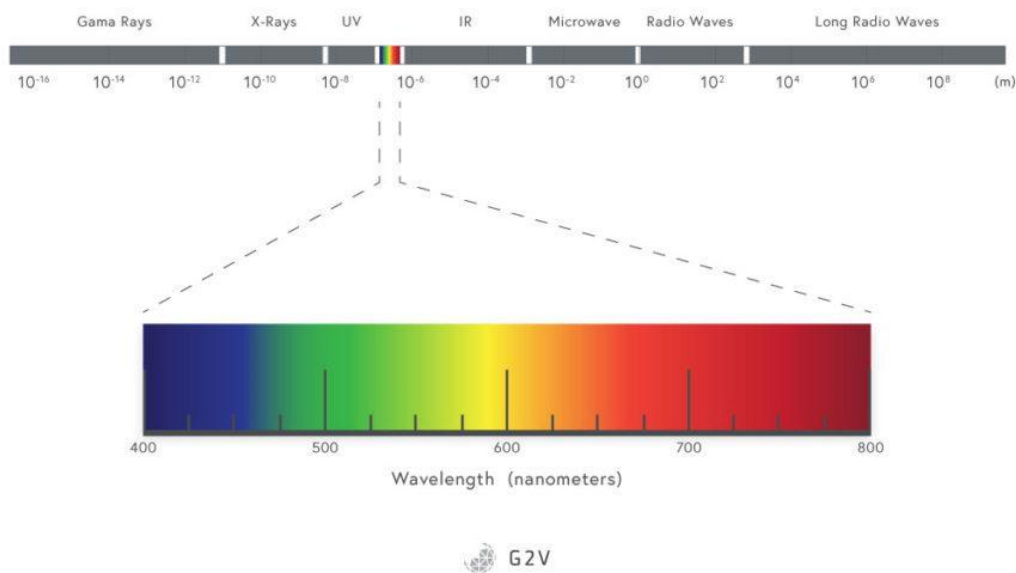


Figure 5 Spectrum of the light

¹⁴ <https://g2voptics.com/solar-simulation/solar-spectrum/>

Although the human eye perceives only the visible portion of the electromagnetic spectrum, plants respond to other solar wavelengths, including the ultraviolet and infrared. This responsiveness is essential to their photochemical processes, through which they generate energy by absorbing wavelengths that extend from the ultraviolet to the infrared¹⁵.

The manner in which these processes operate within a plant may be summarised as follows. When light is absorbed, carbon dioxide and water are converted into essential sugars and oxygen through the action of a pigment known as chlorophyll, a process termed photosynthesis. Chlorophyll absorbs across a broad range of wavelengths but most strongly in the red and blue regions; because it does not absorb green wavelengths, plants appear green to the eye¹⁶. Furthermore, the constituents of a plant reflect, transmit and absorb differing proportions of the near-infrared radiation that is invisible to the human eye. This reflected near-infrared radiation can be detected by satellites and by multispectral or hyperspectral cameras, thereby enabling the study of vegetation. A plant with a higher chlorophyll content reflects more near-infrared radiation than an unhealthy one, and analysing a plant's spectrum of absorption and reflection across the visible and infrared ranges therefore yields valuable information about its health and productivity¹⁷.

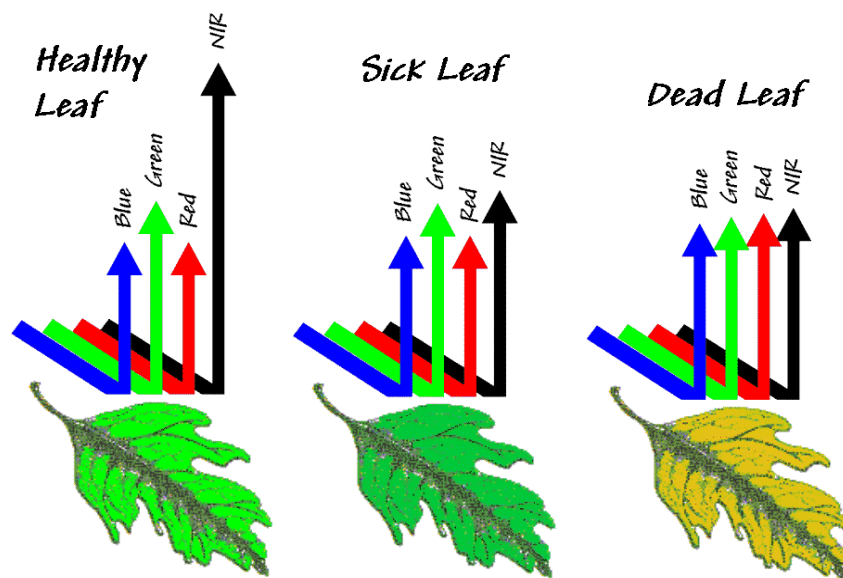


Figure 2 <https://clipart-library.com/clipart/dc486o8xi.htm>

3.2 Multispectral / Hyperspectral Camera

Operating: Whereas a conventional camera records an image using only the three primary colours—red, green and blue—a hyperspectral camera captures a far richer body of data. Equipped with advanced sensors, such cameras decompose the incoming light into numerous spectral bands, gathering information across hundreds of wavelengths, some of which lie beyond human vision, such as the ultraviolet and near-infrared. The remarkable feature of hyperspectral imaging is that it does not merely reproduce a visual representation of a scene but also reveals the material composition of the objects within it, since each pixel of a hyperspectral image constitutes a source of data describing how light interacts with the underlying materials¹⁸.

¹⁵ <https://g2voptics.com/solar-simulation/solar-spectrum>

¹⁶ <https://ubigro.com/light-interaction-with-plants-reflection-emission-and-absorptionlight-interaction-with-plants/>

¹⁷ https://science.nasa.gov/ems/08_nearinfraredwaves/

¹⁸ <https://www.iberdrola.com/about-us/what-we-do/solar-photovoltaic-energy/hyperspectral-cameras>

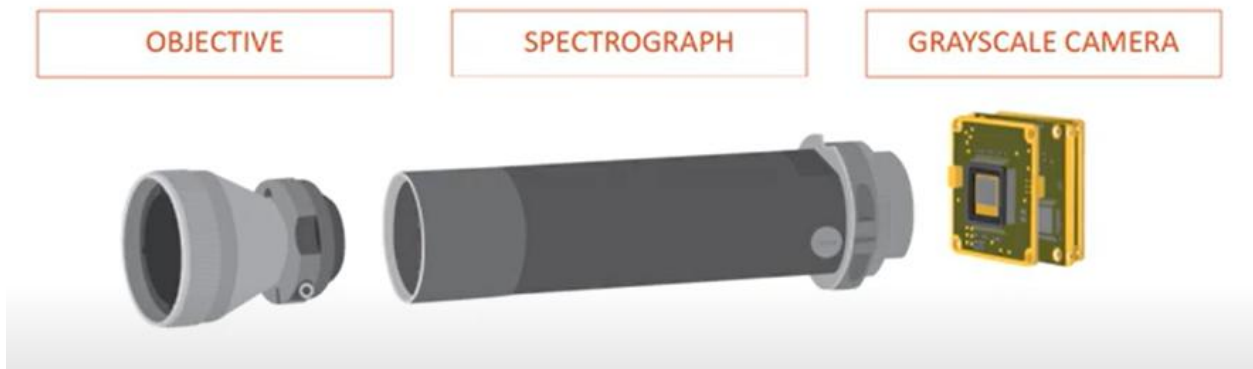


Figure 6 Optical architecture of a hyperspectral imaging camera

3.3 Imaging Spectrograph



Figure 7 Exploded view of an imaging spectrograph

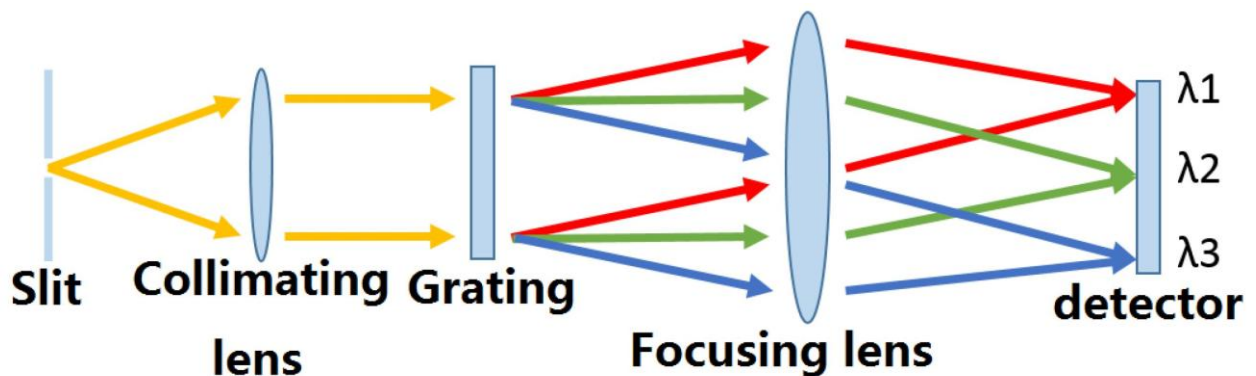


Figure 8 Exploded view of the imaging spectrograph

The instrument comprises an entrance slit, collimating optics, a dispersive element and a focusing lens. The entrance slit restricts the incoming light so that only a single line of the scene is admitted at a time. Because these are line-scan cameras, such a slit is indispensable for the accurate measurement of spectra, and the narrower the slit, the greater the spectral accuracy that can be achieved.

The collimating optics render the light emerging from the slit parallel before it reaches the dispersive element, where the incoming light is separated into its constituent spectrum. The focusing lens is then required to project this dispersed image onto the grayscale sensor.

The central thin circular component is most likely the diffraction grating. It disperses the collimated white light into its constituent wavelengths according to the diffraction equation:

$$m\lambda = d(\sin\theta_i + \sin\theta_d), \text{ where:}$$

- m = diffraction order,
- λ = wavelength,
- d = grating spacing,
- θ_i = incident angle,
- θ_d = diffraction angle.

The last cylindrical component contains the imaging lens. It focuses the separated wavelengths onto the monochrome image sensor.

The greyscale camera measures the intensity of the dispersed light, yielding a spectral image in which each row conveys intensity information for a distinct wavelength. Taken together, the rows constitute the complete spectrum, so that each column contains the full spectrum for a single spatial location, and the image of the entire target is thereby acquired line by line.

The output of a hyperspectral imaging system is often visualized as a hyperspectral data cube. This cube has three dimensions: two spatial dimensions (x and y) corresponding to the scene's layout, and one spectral dimension (z) that represents the range of wavelengths captured¹⁹.

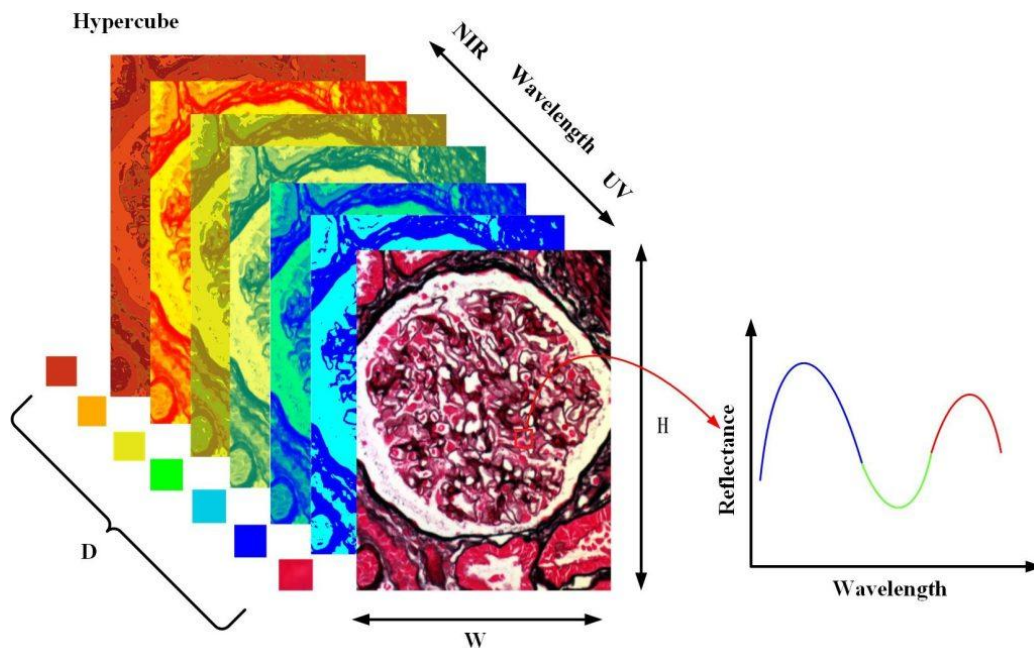


Figure 9 The concept of a hyperspectral data cube (hypercube)

Unlike a conventional colour camera, which records only three broad bands (red, green and blue), a multispectral or hyperspectral sensor captures the reflected light in many narrow wavelength bands. The result is not a single image but a three-dimensional data structure known as a data cube, or hypercube, as illustrated in Figure 9. Two of its dimensions (W and H) are spatial and correspond to the rows and columns of the image, while the third dimension (D) is spectral and contains the successive bands recorded at increasing wavelengths, ranging from the ultraviolet to the near-infrared region of the spectrum.

¹⁹ <https://www.jouav.com/blog/hyperspectral-imaging.html>

Each layer of the cube is therefore a monochromatic image of the same scene, taken at one specific wavelength. Read in the other direction, every individual pixel stores an entire reflectance spectrum, that is, the proportion of light it reflects at each wavelength. This curve, shown on the right-hand side of Figure 9, is called the spectral signature of the pixel. Because different materials reflect and absorb light in characteristic ways, their spectral signatures differ, which makes it possible to distinguish substances that may look identical to the human eye. In the context of precision agriculture, this property is what allows healthy and stressed vegetation to be separated, since living plant tissue reflects strongly in the near-infrared while absorbing in the visible range.

3.4 Field mapping algorithm

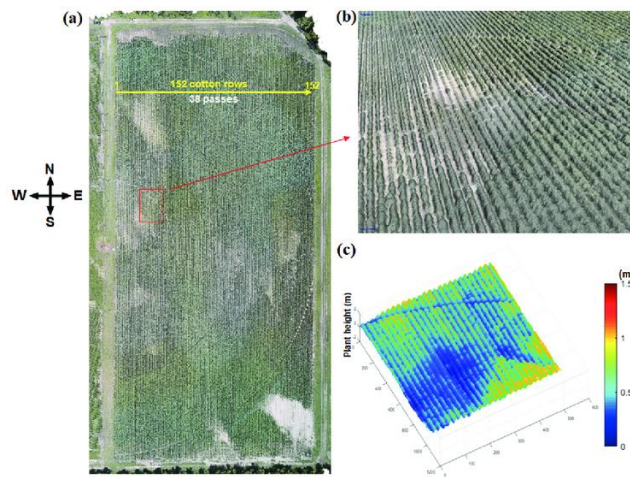


Figure 10 Field mapping algorithm

Field mapping is the procedure through which the many single images taken by a drone during its flight are combined into one geometrically correct representation of the whole field. Because the camera covers only a small surface in each frame, a complete map cannot be produced from a single photograph. For this reason the drone follows a predefined trajectory, usually a parallel "lawnmower" pattern, and records a sequence of images with a high degree of overlap. A forward overlap of about 75 % and a lateral overlap of about 65 % are commonly applied, so that every point on the ground is present in several successive frames.

The mapping algorithm afterwards joins these images by a method called image stitching, which is founded on photogrammetry and computer vision. In the first step, distinctive points, named features or keypoints, are detected in each image with the help of algorithms such as SIFT or ORB.

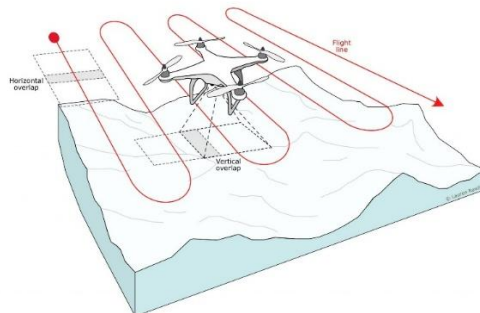


Figure 11 Field mapping algorithm 2

3.5 The SIFT algorithm

The Scale-Invariant Feature Transform (SIFT), proposed by David Lowe in 1999, is still one of the most reliable feature-detection algorithms used in photogrammetric applications. The method detects keypoints by looking for extrema in a so-called scale space. This scale space is built by blurring the image progressively with Gaussian filters and then computing the differences between successive blurred versions, an operation known as the Difference of Gaussians. The blurred image at a given scale is obtained by the convolution of the original image with a Gaussian kernel:

$$L(x,y,\sigma)=G(x,y,\sigma)*I(x,y)$$

$$G(x,y,\sigma)=\frac{1}{2\pi\sigma^2}*e^{-\frac{x^2+y^2}{2\sigma^2}}$$

Where $I(x, y)$ is the input image, G is the Gaussian function and σ controls the amount of blur. The Difference of Gaussians is then defined as the subtraction of two neighboring scales:

$$D(x,y,\sigma)=L(x,y,k\sigma)-L(x,y,\sigma)$$

Where k is a constant multiplicative factor between two consecutive scales. The local extrema of D indicate candidate keypoints.

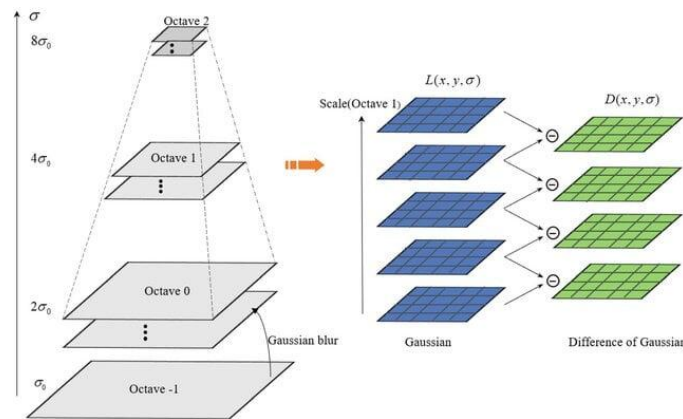


Figure 12 The construction process of the Gaussian pyramid and differential Gaussian pyramid (DOG)²⁰

Figure 12 illustrates in more detail how the scale space of SIFT is organized into octaves. On the left side, the image is represented at several octaves, and within each octave it is blurred by increasing values of σ ; from one octave to the next, the scale parameter is doubled, so that the sequence follows $\sigma_0, 2\sigma_0, 4\sigma_0, 8\sigma_0$, while the image is also sub-sampled²¹. On the right side, the blurred images $L(x,y,\sigma)$ of a single octave are shown, and each pair of neighbouring scales is subtracted in order to obtain the Difference-of-Gaussians images $D(x,y,\sigma)$, represented here in green. In this way, the Gaussian smoothing and the subtraction of consecutive scales are combined into one continuous structure.

After the keypoints have been localised, each of them receives a dominant orientation, which is estimated from the local gradient directions around the point. The gradient magnitude and orientation at each pixel are computed as:

²⁰

²¹ <https://medium.com/@deepanshut041/introduction-to-sift-scale-invariant-feature-transform-65d7f3a72d40>

$$m(x, y) = \sqrt{(L(x + 1, y) - L(x - 1, y))^2 + (L(x, y + 1) - L(x, y - 1))^2}$$

$$\theta(x, y) = \tan^{-1} \left(\frac{L(x, y + 1) - L(x, y - 1)}{L(x + 1, y) - L(x - 1, y)} \right)$$

Finally, a 128-dimensional descriptor is formed from the distribution of the gradients in the neighbourhood of the keypoint. In practice, the region around the point is divided into a 4×4 grid of subregions, and in each subregion a histogram of gradient orientations with 8 bins is calculated, which gives 4×4×8=128 values. The following figure shows this principle, where the small arrows represent the local gradients and the circular diagrams represent the orientation histograms.

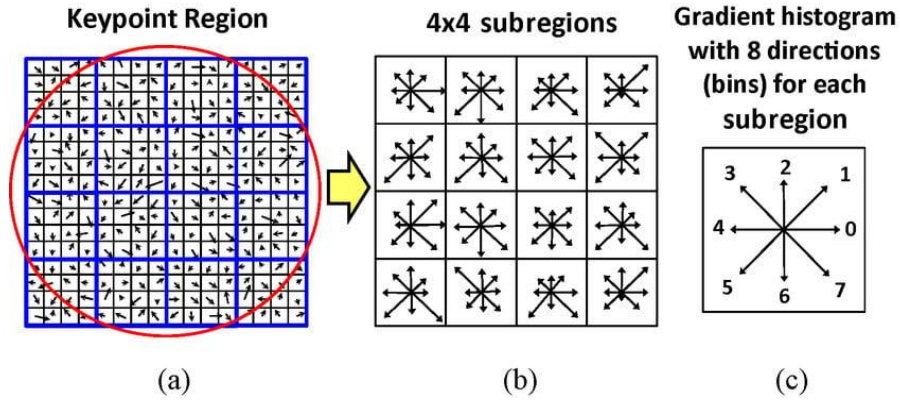


Figure 13 The construction of the SIFT descriptor

Because both the detection and the description stages take the scale and the rotation into account, SIFT features can be matched in a reliable way between images that were taken from different altitudes, from different viewing angles, or under different illumination — exactly the conditions that appear during drone flights. When two descriptors are compared, their similarity is usually measured by the Euclidean distance:

$$d(a, b) = \sqrt{\sum_{i=1}^{128} (a_i - b_i)^2}$$

The main disadvantage of the method is its computational cost. The extensive analysis of the scale space and the high dimension of the descriptors make SIFT relatively slow, which can become a limitation when thousands of aerial images have to be processed.

3.6 The ORB algorithm

ORB (Oriented FAST and Rotated BRIEF), introduced by Rublee and his colleagues in 2011, was designed as a computationally efficient alternative to SIFT. It combines the FAST corner detector with the BRIEF descriptor and adds to both of them the property of rotation invariance.

The FAST detector recognises a keypoint by examining a circle of sixteen pixels around a candidate point p . The central pixel is considered a corner if a sufficient number of contiguous pixels on the circle are all brighter, or all darker, than the centre by more than a threshold t . Each pixel x on the circle is classified according to the rule:

$$S(x) = \begin{cases} \text{darker,} & I_x \leq I_p - t \\ \text{similar,} & I_p - t < I_x < I_p + t \\ \text{brighter,} & I_x \geq I_p + t \end{cases}$$

If at least N consecutive pixels (typically N= 12) belong to the "brighter" or to the "darker" class, the point p is accepted as a corner. The principle is shown in the figure below, where the highlighted arc of pixels around the central point satisfies this condition²².

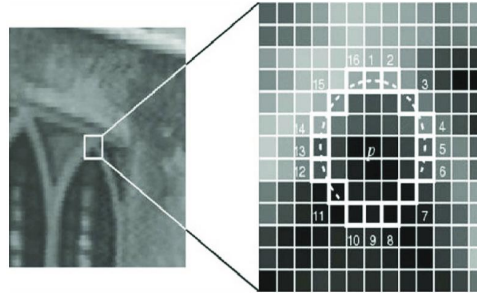


Figure 14 The highlighted arc of pixels

Because the original FAST detector does not provide any orientation, ORB estimates the orientation of each keypoint through the so-called intensity centroid. The moments of the patch are defined as:

$$m_{pq} = \sum_{x,y} x^p y^q I(x,y)$$

For the description stage, ORB uses the BRIEF descriptor, which is a compact binary string generated from simple comparisons between the intensities of pairs of pixels. For a pair of points (x,y) inside the patch, a binary test is defined as:

$$\tau(p; x, y) = \begin{cases} 1, & I(x) < I(y) \\ 0, & \text{otherwise} \end{cases}$$

ORB improves this scheme by rotating the set of test pairs according to the orientation θ computed above, so that the descriptor becomes invariant to rotation, a property that the original BRIEF did not possess.

In practice, the choice between the two algorithms represents a compromise between accuracy and processing speed. SIFT is usually preferred when the quality of the reconstruction is the most important criterion, whereas ORB is more suitable for real-time or resource-limited applications, such as the processing performed on board during the flight.

²² <https://medium.com/thedeephub/detecting-and-tracking-objects-with-orb-using-opencv-d228f4c9054e>

4. Implementation

4.1 Components

For the present project, a quadcopter was constructed from the following components: a DJI F450 frame, four 30 A brushless ESCs, four 2212 920 KV brushless motors, an ICM-20948 sensor, a FlySky FS-i6 transmitter–receiver, an STM32F411CEU “Black Pill” microcontroller and a 3S LiPo battery.

4.1.1 DJI F450 Frame

The DJI F450 is among the most widely used quadcopter airframes in educational contexts. It has a diagonal, motor-to-motor wheelbase of 450 mm, and the frame itself has a mass of approximately 280–300 g, being designed to carry a payload of between 800 g and 1600 g. The lower plate incorporates a power-distribution board (PDB), whose copper traces convey battery power directly to the ESCs. The structure permits the onboard electronics to be upgraded without modification, and the arms are moulded from PA66 reinforced with 30% glass fibre, which affords high stiffness at low mass.

4.1.2 2212 920 KV Brushless DC Motor

This brushless DC (BLDC) motor is highly compatible with the DJI F450 airframe. Fitted with a 10×4.5 propeller and powered by a 3S LiPo battery, each motor produces between 800 and 900 g of thrust; four such motors therefore generate approximately 3.2 kg of thrust. Given that the drone’s total mass lies between 1.0 and 1.3 kg, the resulting thrust-to-weight ratio is sufficient for stable flight.

$$\frac{3200}{1300} = 2.46$$

The motor is rated at 920 KV, where KV denotes the theoretical no-load rotational speed of the motor per volt applied.

$$RPM = KV * V$$

With a 3S battery: $RPM = 920 * 11.1 = 10.200 \text{ RPM}$ This value affords an appropriate balance between rotational speed and torque.

4.1.3 LiPo Battery Selection – GENS ACE G-Tech Soaring 3S 2200 mAh 30C

This is a lithium-polymer (LiPo) battery in a 3S1P configuration, with a nominal voltage of 11.1 V. Its maximum current delivery of 66 A is sufficient for the four motors, whose combined peak draw is between 40 and 60 A.

$$I_{max} = Capacity * C$$

$$I_{max} = 2.2A * 30 = 66A$$

4.1.4 30A Electronic Speed Controller (ESC)

The ESC receives commands from the flight controller in the form of PWM signals, regulates the power drawn from the battery and controls the speed of the brushless motor. It can deliver 30 A continuously, which comfortably exceeds the 10–18 A drawn by a motor at full throttle. It also

incorporates a Battery Eliminator Circuit (BEC) with a 5 V / 2 A output, which is used to power the flight controller, the receiver and the sensor.

4.1.5 FlySky FS-i6 Transmitter + FS-iA6 Receiver

The transmitter is a six-channel, 2.4 GHz radio system operating under the AFHDS 2A (Automatic Frequency-Hopping Digital System) protocol, which provides a stable and interference-resistant link between the pilot and the aircraft. It offers a range of 500–1000 m, employs Gaussian Frequency-Shift Keying (GFSK) modulation and is powered by four AA batteries (6 V). The receiver provides six PWM output channels and accepts an input voltage of 4–6.5 V. Each control stick is mapped to a channel as follows:

CH1 -> Roll,
CH2 -> Pitch,
CH3 -> Throttle,
CH4 -> Yaw,
CH5 -> Mode switch,
CH6 -> Auxiliary.

The receiver is additionally capable of communicating by means of both PWM signals and the IBUS protocol.

4.1.6 ICM-20948 Inertial Measurement Unit (IMU)

The ICM-20948 is a nine-axis Inertial Measurement Unit employed for motion tracking and attitude estimation. It has a VDD operating range of 1.71 V to 3.6 V and integrates three distinct sensing elements within a single chip:

- a three-axis gyroscope, comprising angular-rate sensors for the X-, Y- and Z-axes, with a programmable full-scale range of ± 250 , ± 500 , ± 1000 and ± 2000 dps;
- a three-axis accelerometer, providing digital output for the X-, Y- and Z-axes, with a full-scale range of ± 2 , ± 4 , ± 8 and ± 16 g;
- 3-axis magnetometer:
 - silicon monolithic Hall-effect sensors with a magnetic concentrator;
 - an output data resolution of sixteen bits;
 - a full-scale measurement range of ± 4900 μ T.

The ICM-20948 can be addressed either over I²C, at up to 100 kHz in standard mode or up to 400 kHz in fast mode, or over SPI at up to 7 MHz. Rather than placing all of its registers within a single address space, the device organises them into four user banks (0–3), the active bank being selected through the REG_BANK_SEL (0x7F) register.

Table 1 User Banks in ICM

Bank	Main Purpose
User Bank 0	Sensor data, interrupts, FIFO, power management
User Bank 1	Self-test and offset calibration registers
User Bank 2	Gyroscope and accelerometer configuration
User Bank 3	I ² C master configuration (used to communicate with the internal magnetometer)

4.1.7 STM32F411CEU Black Pill

Table 2 Comparison the three microcontrollers that I used

Feature	Arduino Uno	STM32F103C8T6 (“Blue Pill”)	STM32F411CEU (“Black Pill”)
CPU	ATmega328P	ARM Cortex-M3	ARM Cortex-M4
Clock speed	16 MHz	72 MHz	100 MHz
FPU (floating point)	✗ No	✗ No	✓ Yes
RAM	2 KB	20 KB	128 KB
Flash	32 KB	64 KB	512 KB
Real-time performance	Low	Medium	High
PID computation ability	Limited	OK	Excellent
Sensor fusion suitability	Poor	Basic	Strong
UART/SPI/I2C support	Limited	Good	Excellent
PWM precision	Low	Medium	High

This flight controller is based on an ARM Cortex-M4 core running at 100 MHz and equipped with a hardware floating-point unit (FPU), which permits control algorithms to be executed efficiently. It offers a variety of communication interfaces, including SPI, UART and timer peripherals, thereby allowing the IMU, the GPS module and the radio-control receiver to be integrated seamlessly.

A comparative analysis was undertaken of three microcontroller platforms: the Arduino Uno (ATmega328P), STM32F103C8T6, and STM32F411CEU. The evaluation was based on computational performance, memory capacity, peripheral availability, and suitability for real-time UAV control systems. The Arduino Uno, while widely used in educational applications, lacks sufficient processing power and floating-point support for high-frequency flight control algorithms. The STM32F103C8T6 offers improved performance and is commonly used in basic UAV systems; however, its lack of a hardware floating-point unit limits its efficiency in sensor fusion and advanced control computations. In contrast, the STM32F411CEU provides significantly higher processing capability, expanded memory resources, and a hardware FPU, enabling efficient execution of PID control loops and real-time sensor fusion algorithms. Based on this analysis, the STM32F411CEU was selected as the optimal platform for implementing a custom UAV flight controller.

4.2 Programming

4.2.1 Programming environment

The firmware was developed within the STM32CubeIDE environment.

Table 3 Comparison the two programming environment

Feature	Arduino IDE	STM32CubeIDE
Hardware Access	High-level libraries	Direct peripheral configuration
Register-Level Programming	Limited	Full support
Debugging	Basic Serial Monitor	Integrated debugger (ST-LINK)
Peripheral Configuration	Manual libraries	CubeMX graphical configuration
RTOS Support	Limited	Native FreeRTOS support
Optimization	Basic	Advanced compiler optimization
UAV Suitability	Small projects	Professional UAV systems

Although Arduino IDE simplifies embedded development through high-level abstraction libraries, it provides limited control over low-level hardware resources required for real-time flight control applications. Programming the drone requires precise control over hardware peripherals, including timers, interrupts, SPI communication, UART interfaces, and PWM signal generation. STM32CubeIDE provides direct access to these resources while also integrating advanced debugging tools and the CubeMX peripheral configuration framework. These capabilities facilitate the implementation and testing of real-time control algorithms and sensor interfaces.

The firmware development proceeded through the following stages:

1. Motor control
2. IMU integration
3. Attitude estimation
4. Proportional–Integral–Derivative (PID) control
5. RC joystick integration

4.2.2 Motor Control

Before the motor can be controlled, it is essential to understand how timer interrupts operate on STM32 microcontrollers, together with the mechanisms that underlie them.

Appendix 1 illustrates the operating principle of this timer. In the case shown, the auto-reload value is set to 36 and the timer counts upwards; when the counter reaches 36, a counter-overflow event occurs and the counter is reset to zero. This overflow takes place at a constant period, which is determined by the timer's clock frequency and by the auto-reload value²³.

To generate a timer interrupt, the timer must first be configured in the CubeMX tool. For this purpose TIM4 was selected, and its prescaler and period were set accordingly, since these parameters directly determine the frequency of the timer interrupt. The function of the repetition

²³ <https://www.steppeschool.com/blog/stm32-timer-interrupt>

counter is to control the rate of the timer update interrupt: if its value is N, the update interrupt is invoked once for every N+1 timer overflows.

$$\text{Interrupt frequency} = \frac{\text{Clock frequency}}{(\text{prescaler} + 1)(\text{AutoReload} + 1)(\text{RepetitionCounter} + 1)}$$

In the present configuration the STM32 clock frequency is 96 MHz, the prescaler is 95, the auto-reload (timer period) is 9999 and the repetition counter is zero, which yields an interrupt frequency of 100 Hz.

Appendix 2 shows that a PWM signal is characterised by two quantities: its frequency (or period) and its duty cycle. The duty cycle is the ratio between the pulse width and signal period. By modulating the duty cycle, the power delivered by the signal can be controlled.

Appendix 3 illustrates how the microcontroller generates PWM signals. The timer begins at zero and counts upwards; while the counter remains below the Capture/Compare register (CCRx), the PWM signal is held high, after which it switches low and remains low until the counter reaches the auto-reload value. The frequency of the PWM signal can thus be adjusted through the auto-reload value, while the CCRx register governs the duty cycle.

4.2.3 IMU Integration

Appendix 4 illustrates the communication between the STM32 and the ICM-20948, for which the SPI (Serial Peripheral Interface) protocol was used. *Appendix 5* presents an overview of the SPI interface. Connecting the sensor to the STM32 over SPI requires the following lines²⁴:

SCLK -> Serial Clock

MISO -> Master-in-Slave-out

MOSI -> Master-out-Slave-in

CS -> Chip Select

Appendix 6 illustrates how the master transmits data. The SPI master initiates communication by driving the Chip Select line low, thereby activating the target slave. Transmission on the MOSI line then proceeds in two phases: the master first sends a write command together with the destination register address, indicating to the slave where the data are to be stored, and subsequently transmits the actual data bytes. Throughout this exchange the master supplies a continuous clock signal on the SCLK line, which synchronises the transfer of each bit. Because a write operation is in progress, the slave returns no data and the MISO line remains idle; once all data have been transmitted, the master returns CS to the high state to signal the end of the transaction.

Appendix 7 illustrates how the master receives data. The SPI master again begins by driving the Chip Select line low in order to activate the target slave, after which the exchange proceeds in two phases. In the first phase the master transmits, on the MOSI line, a read command together with the register address, informing the slave which register is to be read, while the MISO line remains idle. In the second phase the master sends a dummy byte (0xFF) on MOSI so as to generate the requisite SCLK pulses, and the slave uses these pulses to shift the requested data back to the master over the MISO line. Once the data have been returned, the master restores the CS line to the high state.

Appendix 8 illustrates the manner in which SPI supports multi-byte read operations within a single Chip Select transaction, an approach that is more efficient than performing a series of separate

²⁴ https://digilent.com/blog/what-is-spi/?srsltid=AfmBOoq3_XEKdI7e5IZEHDu4_BQanq7qfjNXiLxMreaT4YuN_-Jg-fgr

single-byte reads. The master initiates the transfer by driving CS low and sending a read command together with the starting register address, in which the most significant bit denotes a read operation.

Clock polarity (CPOL) and Clock Phase (CPHA)

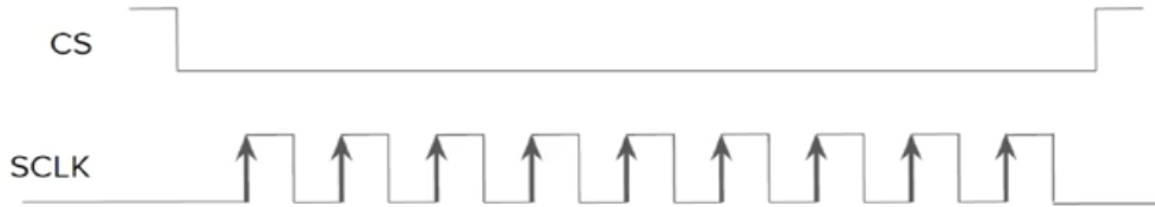


Figure 15 CPOL= 0 and CPHA = 0: clock is low in idle state and sampling on the leading clock edge



Figure 16 CPOL = 0 and CPHA = 1: clock is low in idle state and sampling on the second clock edge

The CPOL and CPHA bits together define four distinct SPI modes, and the master and slave must be configured to use the same mode if communication is to succeed.

4.2.4 Block Diagram

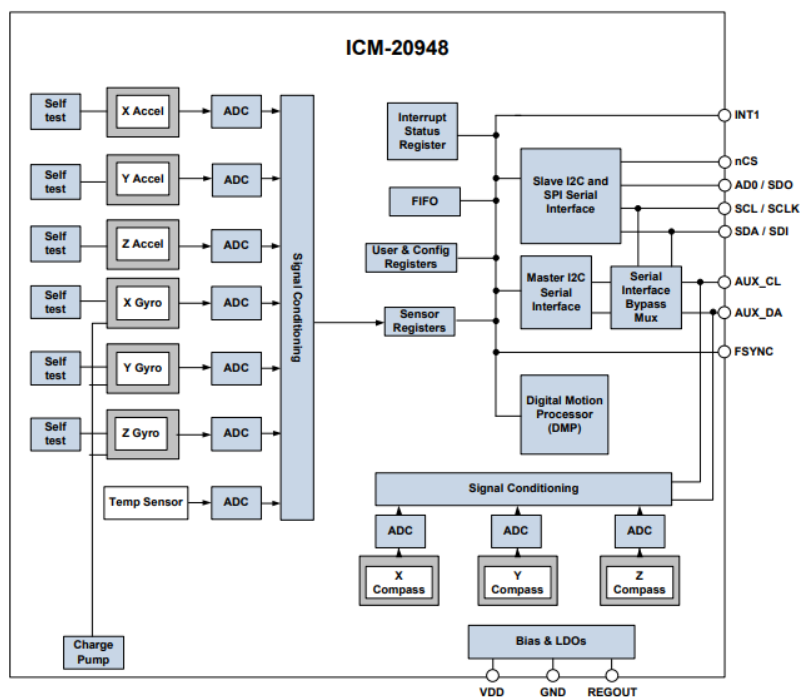


Figure 17 ICM-20948 Block Diagram

The values produced by the gyroscope and accelerometer are stored directly in the sensor registers. Obtaining data from the magnetometer, however, is less straightforward, because the magnetometer is not fully integrated with the main sensing core; instead, it is connected internally by means of an I²C master interface. The magnetometer therefore requires this master I²C interface to be configured, whereupon the compass data are read and likewise stored in the sensor registers, and all of the data are ultimately transmitted over the SPI interface. This I²C master facility also makes it possible to attach a further sensor to the ICM-20948 and to read data from it.

Table 4 Register table for Magnetometer

NAME	ADDRESS	READ/WRITE	DESCRIPTION	BIT WIDTH	EXPLANATION
WIA2	01H	READ	Device ID	8	
ST1	10H	READ	Status 1	8	Data status
HXL	11H	READ	Measurement data	8	X-axis data
HXH	12H			8	
HYL	13H			8	Y-axis data
HYH	14H			8	
HZL	15H			8	Z-axis data
HZH	16H			8	
ST2	18H	READ	Status 2	8	Data status
CNTL2	31H	READ/ WRITE	Control 2	8	Control Settings
CNTL3	32H	READ/ WRITE	Control 3	8	Control Settings
TS1	33H	READ/ WRITE	Test	8	DO NOT ACCESS
TS2	34H	READ/ WRITE	Test	8	DO NOT ACCESS

Beginning at bit 11, six bytes must be read: two bytes for each of the X, Y and Z axes. As with the gyroscope and accelerometer, the measurement for each axis is therefore represented as a sixteen-bit signed integer. For this sensor it is also necessary to read the 18th bit, since otherwise the measurement data are not refreshed.

Table 5 Control 3 register

ADDR	REGISTER NAME	D7	D6	D5	D4	D3	D2	D1	D0
READ/WRITE REGISTER									
32H	CNTL3	0	0	0	0	0	0	0	SRST
Reset		0	0	0	0	0	0	0	0

This register is located at address 32. By means of this magnetometer control register, all magnetometer registers can be reset by setting the SRST bit to 1; following the reset, the SRST bit reverts automatically to 0.

Table 6 Control 2 register

ADDR	REGISTER NAME	D7	D6	D5	D4	D3	D2	D1	D0
READ/WRITE REGISTER									
31H	CNTL2	0	0	0	MPDE4	MODE3	MODE2	MODE1	MODE0
Reset		0	0	0	0	0	0	0	0

The operating mode is selected by configuring this register, as follows:

“00000”: Power-down mode

“00001”: Single measurement mode

“00010”: Continuous measurement mode 1

“00100”: Continuous measurement mode 2

“00110”: Continuous measurement mode 3

“01000”: Continuous measurement mode 4

“10000”: Self-test mode

In the present implementation, continuous measurement mode 4 was used, in which the sensor samples at 100 Hz and stores the resulting data in the HXL to HZH registers.

In embedded systems, transferring data from a peripheral such as an SPI sensor to memory can be done in two ways. In blocking mode, the CPU is directly responsible for reading each incoming byte from the peripheral and storing it in memory. During this process, the CPU is fully occupied and cannot execute any other instructions, which wastes processing time and reduces system responsiveness. *Appendix 9* illustrates that the solution to this problem is Direct Memory Access (DMA). DMA is a dedicated hardware unit that can transfer data autonomously between a peripheral and memory — or between two memory locations — without any CPU intervention. The CPU only needs to configure the DMA controller once with the source address, destination address, and transfer size. The DMA then handles the entire data movement independently over the system bus. Once the transfer is complete, the DMA notifies the CPU via an interrupt. This approach frees the CPU to perform other computations in parallel, significantly improving system efficiency and real-time performance.

4.2.5 Attitude Estimation

Before the accelerometer data can be used in the orientation estimation algorithm, it must be normalized. First, the magnitude of the raw acceleration vector is computed as the Euclidean norm of its three components:

$$\text{accel_mag} = \sqrt{\text{accel_x}^2 + \text{accel_y}^2 + \text{accel_z}^2}$$

Each axis component is then divided by this magnitude, yielding a normalized unit vector [ax, ay, az]

$$\begin{bmatrix} \text{accel_x} \\ \text{accel_y} \\ \text{accel_z} \end{bmatrix} \rightarrow \begin{bmatrix} \text{accel_x}/\text{accel_mag} \\ \text{accel_y}/\text{accel_mag} \\ \text{accel_z}/\text{accel_mag} \end{bmatrix} = \begin{bmatrix} \text{ax} \\ \text{ay} \\ \text{az} \end{bmatrix}$$

The key property of this normalization is that the resulting vector always satisfies $\sqrt{\text{ax}^2 + \text{ay}^2 + \text{az}^2} = 1$ meaning its magnitude is exactly 1 regardless of the raw sensor values. This ensures that the accelerometer contributes only directional information — specifically the direction of gravity — to the filter, without being affected by scaling differences or sensor gain.

The same normalization procedure is applied to the magnetometer data.

$$\text{compass_mag} = \sqrt{\text{mag_x}^2 + \text{mag_y}^2 + \text{mag_z}^2}$$

$$\begin{bmatrix} \text{mag_x} \\ \text{mag_y} \\ \text{mag_z} \end{bmatrix} \rightarrow \begin{bmatrix} \text{mag_x}/\text{compass_mag} \\ \text{mag_y}/\text{compass_mag} \\ \text{mag_z}/\text{compass_mag} \end{bmatrix} = \begin{bmatrix} \text{mx} \\ \text{my} \\ \text{mz} \end{bmatrix}$$

Gyroscope scaling: to understand why scaling is necessary, it should be noted that the gyroscope outputs raw sixteen-bit integer values ranging from $[-2^{15} + 1; 2^{15} - 1]$ from its ADC, which corresponds to the configured full-scaled angular rate range. Before this data can be used in an orientation estimation algorithm, it must be scaled to physical units — specifically radians per

second (rad/s). This is done by applying a scaling factor that combines two conversions into a single multiplication.

$$Scale_factor = \frac{\pi}{180} * \frac{full_scale}{32767}$$

Each raw gyroscope axis value is multiplied by this factor to produce the calibrated angular velocity components gx, gy, and gz in rad/s, which are then passed directly to the orientation filter.

$$\begin{bmatrix} gyro_x \\ gyro_y \\ gyro_z \end{bmatrix} \rightarrow \begin{bmatrix} gyro_x \times Scale_factor \\ gyro_y \times Scale_factor \\ gyro_z \times Scale_factor \end{bmatrix} = \begin{bmatrix} gx \\ gy \\ gz \end{bmatrix}$$

An important consideration when working with the ICM-20948 is that the accelerometer/gyroscope and the magnetometer do not share the same coordinate frame. The accelerometer and gyroscope axes follow a standard right-hand orientation +Z pointing upward out of the chip surface (Figure 15).

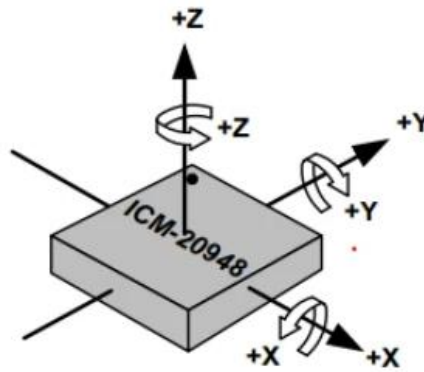


Figure 18 Orientation of Axes of Sensitivity and Polarity of Rotation

However, the magnetometer, which is a separate die integrated into the same package, has a different axis orientation: its +Z axis points downward, which is opposite to that of the accelerometer and gyroscope (Figure 15). This means that raw magnetometer readings cannot be directly combined with accelerometer and gyroscope data in a sensor fusion algorithm without first applying an axis remapping transformation. Therefore, before passing magnetometer data to the orientation filter, the appropriate axis signs must be corrected so that all three sensors share a common reference frame.

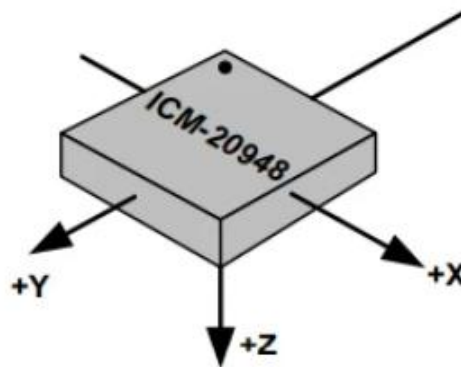


Figure 19 Orientation of Axes of Sensitivity for Magnetometer

Body frame vs. Global frame

Appendix 10 represents the difference between global frame and body frame. The global frame defines the reference frame, which never moves, it is fixed to the Earth. No matter if the object rotates or translates, the global frame remains the same. This is the NED convention, where x points to north, y points to east, z points to earth center.

The body frame, by contrast, moves with the object: as the aircraft rolls, pitches or yaws, the body frame rotates with it. In this convention the x-axis points forward, the y-axis points to the left and the z-axis points downward. The task is thus to establish the relationship between the global and the body frame.

Appendix 11 represents the core concept, that any arbitrary 3D rotation of a rigid body can be decomposed into three successive simple rotations, each around one axis. These three angles together are called Euler angles.

Table 7 Difference between roll, pitch, yaw

Symbol	Name	Axis	Motion (aircraft analogy)
ϕ (phi)	Roll	X-axis	Tilting left/right — like a barrel roll
θ (theta)	Pitch	Y-axis	Nose up/down — like climbing or diving
ψ (psi)	Yaw	Z-axis	Turning left/right — like steering

Euler angles are not commutative — the order in which you apply the three rotations changes the result. There are 12 possible valid sequences, for aerospace and IMU applications the **ZYX sequence (z-y-x)** is standard because it naturally maps to yaw → pitch → roll, matching how aircraft and drones move. Euler angles have one major weakness — gimbal lock. When the pitch angle reaches $\pm 90^\circ$, the roll and yaw axes align, and one degree of freedom is lost.

Rotation matrices for each yaw-pitch roll angles:

$$R_{yaw} = \begin{bmatrix} \cos(\psi) & \sin(\psi) & 0 \\ -\sin(\psi) & \cos(\psi) & 0 \\ 0 & 0 & 1 \end{bmatrix} \text{ yaw angle}$$

$$R_{pitch} = \begin{bmatrix} \cos(\theta) & 0 & -\sin(\theta) \\ 0 & 1 & 0 \\ \sin(\theta) & 0 & \cos(\theta) \end{bmatrix} \text{ pitch angle}$$

$$R_{roll} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos(\phi) & \sin(\phi) \\ 0 & -\sin(\phi) & \cos(\phi) \end{bmatrix} \text{ roll angle}$$

By multiplying roll-pitch-yaw rotation matrices, we can define full rotation matrix. Frame rotation:

$$R_{total} = R_{roll} * R_{pitch} * R_{yaw} = \begin{bmatrix} c(\psi)c(\theta) & s(\psi)c(\theta) & -s(\theta) \\ c(\psi)s(\theta)s(\phi) - c(\phi)s(\psi) & c(\phi)c(\psi) + s(\phi)s(\psi)s(\theta) & c(\theta)s(\phi) \\ s(\phi)s(\psi) + c(\phi)c(\psi)s(\theta) & c(\psi)s(\theta)s(\phi) - s(\phi)c(\psi) & c(\theta)c(\phi) \end{bmatrix}$$

Knowing roll-pitch-yaw angles, we can map vectors (points) from one frame to another. When defining orientation using Euler angles, we have to identify Euler angles using the IMU data. Frame rotation:

$$\begin{bmatrix} x_{body} \\ y_{body} \\ z_{body} \end{bmatrix} = R_{total} * \begin{bmatrix} x_{earth} \\ y_{earth} \\ z_{earth} \end{bmatrix}$$

4.3 Accelerometer and Euler Angles

It is possible to extract Roll and Pitch angles directly from the accelerometer by linking the body frame measurements to the Earth frame through the rotation matrix.

In general, the accelerometer measures specific force – actual acceleration minus gravity:

$$measurement = [ax_{actual} - gx \ ay_{actual} - gy \ az_{actual} - gz]^T$$

After normalization, if we assume the object is **not accelerating** (static or slow motion), the actual acceleration terms drop out, leaving only the gravity component:

$$[gx \ gy \ gz]^T = [ax \ ay \ az]^T$$

This means the normalized accelerometer vector **is** the gravity vector expressed in the body frame.

In the Earth/global frame, gravity points purely downward along Z:

$$gravity \ in \ Earth \ frame = [0 \ 0 \ 1]^T$$

$$[ax \ ay \ az]^T = R_{total} * [0 \ 0 \ 1]^T$$

Multiplying R_{total} by $[0, 0, 1]^T$ simply extracts the **third column** of the rotation matrix:

$$\begin{bmatrix} ax \\ ay \\ az \end{bmatrix} = \begin{bmatrix} -s(\theta) \\ c(\theta)s(\phi) \\ c(\theta)c(\phi) \end{bmatrix}$$

From $ax = -\sin(\theta)$ can be calculated $\theta = \arcsin(-ax)$ and it is possible to get the pitch angle.

From $ay = \cos(\theta)\sin(\phi)$ can be calculated $\phi = \arctan(\frac{ay}{az})$ and it is to get the roll angle.

Yaw angle cannot be obtained from the accelerometer because the gravity is symmetric around the Z-axis, so rotating around it doesn't change $[ax, ay, az]$ at all. Yaw requires the magnetometer.

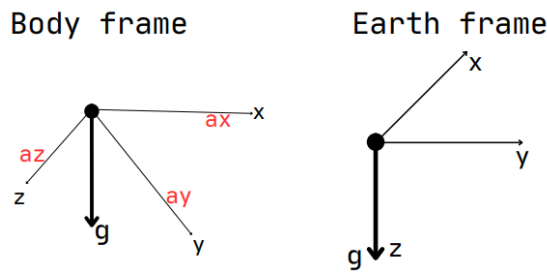


Figure 20 Gravity in body and earth frame

From ay and az it is easy to get the value of $\cos(\theta)$

$$ay^2 + az^2 = \cos^2(\theta)(\sin^2(\phi) + \cos^2(\phi)) = \cos^2(\theta)$$

$$\cos(\theta) = \pm\sqrt{ay^2 + az^2}$$

Table 8 Atan2 function

Case	Roll ϕ	Pitch θ
$\cos \theta > 0$ (normal)	$\text{atan2}(ay/\sqrt{ay^2+az^2}, az/\sqrt{ay^2+az^2})$	$\text{atan2}(-ax, \sqrt{ay^2+az^2})$
$\cos \theta < 0$ (upside down)	$\text{atan2}(-ay/\sqrt{ay^2+az^2}, -az/\sqrt{ay^2+az^2})$	$\text{atan2}(-ax, -\sqrt{ay^2+az^2})$

The two solutions handle the case when the sensor is flipped upside down (negative $\cos \theta$).

The use of atan2 rather than the standard atan function is essential here: while atan receives only a ratio and cannot distinguish between two angles that differ by 180° , atan2 accepts both the numerator and denominator as separate arguments and uses their individual signs to correctly determine the angle across all four quadrants of the unit circle, covering the full -180° to $+180^\circ$ range.

$$\text{atan2}(-x1, -y1) = \pi + \alpha$$

$$\text{atan2}(x1, y1) = \alpha$$

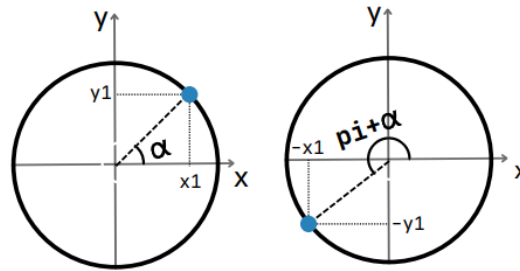


Figure 21 The reason for using atan2

Just as the accelerometer gave us **Roll and Pitch** by relating body-frame gravity to Earth-frame gravity, the magnetometer gives us **Yaw** by relating body-frame magnetic flux to Earth-frame magnetic flux.

In the Earth frame, the magnetic field vector has **no East (Y) component** — it lies only in the X-Z plane (pointing toward North and slightly downward):

$$\text{Earth frame} = \begin{bmatrix} M_x & 0 & M_z \end{bmatrix}^T$$

The magnetometer measures this same field rotated into the body frame:

$$\begin{bmatrix} m_x \\ m_y \\ m_z \end{bmatrix} = R_{total} * \begin{bmatrix} M_x \\ 0 \\ M_z \end{bmatrix}$$

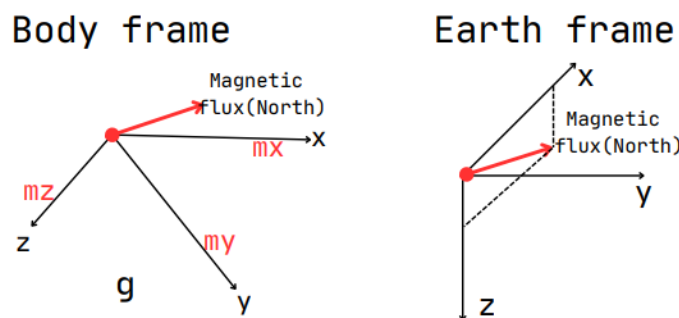


Figure 22 Magnetic flux in body and earth frame

The gyroscope measures angular velocity along the three body-frame axes, producing readings denoted $[p, q, r]$. However, these body-frame angular rates do not correspond directly to the rates of change of the Euler angles $[\phi, \theta, \psi]$, because each Euler angle is defined with respect to a different intermediate frame.

The total body angular velocity is built up from three contributions applied **sequentially** in the ZYX order:

$$\begin{bmatrix} p \\ q \\ r \end{bmatrix} = \begin{bmatrix} \dot{\phi} \\ 0 \\ 0 \end{bmatrix} + R_{roll} \begin{bmatrix} 0 \\ \dot{\theta} \\ 0 \end{bmatrix} + R_{roll} * R_{pitch} \begin{bmatrix} 0 \\ 0 \\ \dot{\psi} \end{bmatrix}$$

$$\begin{bmatrix} p \\ q \\ r \end{bmatrix} = \begin{bmatrix} 1 & 0 & -s(\theta) \\ 0 & c(\phi) & c(\theta)s(\phi) \\ 0 & -s(\phi) & c(\theta)s(\phi) \end{bmatrix} * \begin{bmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{bmatrix}$$

if we inverse the equation

$$\begin{bmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{bmatrix} = \begin{bmatrix} 1 & s(\theta)\tan(\theta) & c(\theta)\tan(\theta) \\ 0 & c(\phi) & -s(\phi) \\ 0 & s(\phi)/c(\theta) & c(\phi)/c(\theta) \end{bmatrix} * \begin{bmatrix} p \\ q \\ r \end{bmatrix}$$

$$\begin{bmatrix} \dot{\phi}_{new} \\ \dot{\theta}_{new} \\ \dot{\psi}_{new} \end{bmatrix} = \begin{bmatrix} 1 & s(\theta)\tan(\theta) & c(\theta)\tan(\theta) \\ 0 & c(\phi) & -s(\phi) \\ 0 & s(\phi)/c(\theta) & c(\phi)/c(\theta) \end{bmatrix} * \begin{bmatrix} p\Delta t \\ q\Delta t \\ r\Delta t \end{bmatrix} + \begin{bmatrix} \dot{\phi}_{old} \\ \dot{\theta}_{old} \\ \dot{\psi}_{old} \end{bmatrix}$$

$$[p\Delta t \ q\Delta t \ r\Delta t]^T = [g_x \ g_y \ g_z]^T$$

The total body angular velocity is the sum of three contributions: the roll rate acts directly on the body X-axis, while the pitch and yaw rates must first be transformed through the appropriate partial rotation matrices before being added.

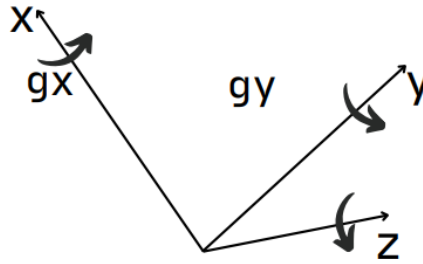


Figure 23 Gyroscope illustration

There are two fundamental problems, accelerometer and magnetometer give absolute orientation referenced to gravity and North, but they are sensitive to vibrations, linear accelerations and magnetic interference. The signal oscillates rapidly around the true value, which is good for long-term accuracy, but it is bad for short-term smoothness. The gyroscope gives very smooth, fast angular velocity readings, but it integrates small errors over time, thus the angle slowly wanders away from truth. Even at rest, tiny bias in the gyro accumulates thus the estimated angle keeps changing. It is good for short-term dynamics, bad for long-term accuracy. For this there is a solution which is combining them with a complementary filter, use gyro for fast dynamics, accel/mag to correct long term drift.

$$\begin{bmatrix} \dot{\phi}_{comp} \\ \dot{\theta}_{comp} \\ \dot{\psi}_{comp} \end{bmatrix} = \begin{bmatrix} \dot{\phi}_{gyro} \\ \dot{\theta}_{gyro} \\ \dot{\psi}_{gyro} \end{bmatrix} + (1 - ALPHA) * \begin{bmatrix} \dot{\phi}_{acc_mag} - \dot{\phi}_{gyro} \\ \dot{\theta}_{acc_mag} - \dot{\theta}_{gyro} \\ \dot{\psi}_{acc_mag} - \dot{\psi}_{gyro} \end{bmatrix}$$

When pitch = 90° (nose pointing straight up), the roll axis and yaw axis **align** with each other. Rotating around either produces the same motion — the system loses one independent axis of rotation.

4.3.1 Quaternion

To overcome the singularity of Euler angles caused by gimbal lock, orientation is represented using quaternions. A quaternion is a four-component hypercomplex number of the form:

$$q = s + xi + yj + zk, \text{ where } s, x, y, z \in \mathbb{R}$$

The imaginary units follow special multiplication rules:

$$i^2 = j^2 = k^2 = -1$$

$$ij = k, jk = i, ki = j$$

$$ji = -k, kj = -i, ik = -j$$

Compactly written as:

$$q = [s, v] \text{ } s \in \mathbb{R}, v \in \mathbb{R}^3$$

For orientation purposes, only unit quaternions are used, meaning their norm $\sqrt{s^2 + x^2 + y^2 + z^2} = 1$. The conjugate of a quaternion is obtained by negating the vector part: $q^* = [s, -v]$, and for a unit quaternion the inverse equals the conjugate: $q^{-1} = q^*$. A 3D point $p = (x_p, y_p, z_p)$ is represented as a pure quaternion $p = [0, x_p, y_p, z_p]$. A rotation by angle θ around a unit axis $\hat{v} = (x, y, z)$ is encoded as the rotation quaternion $q = [\cos(\theta/2), \sin(\theta/2)x, \sin(\theta/2)y, \sin(\theta/2)z]$. To rotate a vector, the sandwich product $p' = qpq^{-1}$ is applied; to perform a frame rotation, the inverse product $p' = q^{-1}pq$ is used instead. Any rotation quaternion can be converted to an equivalent 3×3 rotation matrix $R(q)$, providing a direct link between the quaternion representation and the rotation matrices used in the body-to-global frame transformations derived earlier. Sequential rotations are composed by quaternion multiplication: for frame rotations, n successive rotations defined by quaternions q_1 through q_n combine as $q_1q_2\dots q_n$, enabling the complete orientation to be built incrementally from each sensor update. The key advantage of quaternions over Euler angles is that they represent any 3D orientation continuously and without singularity, making them the standard choice for real-time orientation estimation in embedded systems.

4.3.2 Open-Loop vs Closed-Loop Control

Motor control can be implemented in two ways: open-loop or closed-loop. In open-loop control, the controller sends a fixed command to the plant — for example a fixed PWM duty cycle — without measuring the actual output. While simple to implement, open-loop control cannot compensate for disturbances, load variations, or mechanical imperfections, meaning the motor may not reach or maintain the desired speed or position.

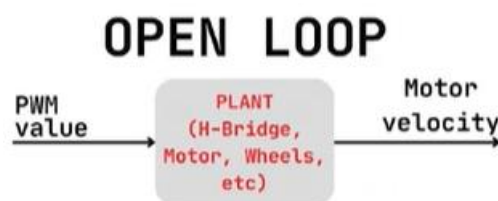


Figure 24 Open loop

Closed-loop control solves this by continuously measuring the actual output via a sensor such as an encoder, computing the error between the reference and the measured value, and dynamically adjusting the control signal to drive the error toward zero. This feedback mechanism makes the system self-correcting and robust to real-world disturbances.

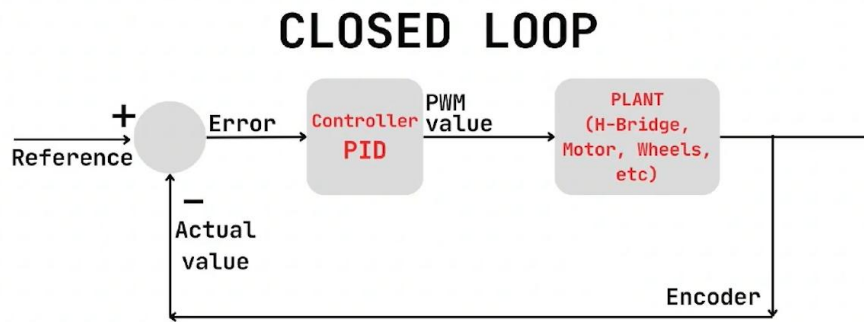


Figure 25 Closed loop

The control algorithm applied is the PID (Proportional-Integral-Derivative) controller, which output $u(t)$ is computed as the weighted sum of three terms. The three components:

$$e(t) = r(t) - y(t) (\text{error} = \text{reference} - \text{actual})$$

$$u(t) = K_p * e(t) + K_i * \int e(t)dt + K_d * \frac{de(t)}{dt}$$

The proportional term $K_p \times e(t)$ produces a correction proportional to the current error and determines the speed of response. The integral term $K_i \times \int e(t)dt$ accumulates the error over time and eliminates steady-state offset that the proportional term alone cannot correct. The derivative term $K_d \times de(t)/dt$ responds to the rate of change of the error and acts as a damper to reduce overshoot and oscillation. In the discrete-time implementation running on the STM32 microcontroller, the integral is approximated by accumulating the error sum over successive time steps multiplied by the sampling period T_s , while the derivative is approximated by the difference between the current and previous error divided by T_s . The gains K_p , K_i , and K_d are tuning parameters that must be selected to achieve the desired balance between response speed, stability, and steady-state accuracy for the specific plant being controlled.

4.4 Joystick integration

The receiver decodes the RF signal and forwards the channel data to the STM32 flight controller via the IBUS protocol. IBUS (Intelligent Bus) is a digital serial protocol that transmits all channel values over a single UART wire at 115200 baud, replacing the traditional approach of using one PWM wire per channel. Each IBUS packet consists of 32 bytes transmitted every 7 milliseconds, yielding an effective update rate of approximately 140 Hz. The packet begins with two synchronization bytes (0x20 and 0x40), followed by up to 14 channel values each encoded as a 16-bit little-endian integer, and concludes with a 16-bit checksum computed as the one's complement sum of all preceding bytes, which the STM32 verifies to detect and discard corrupted packets. Each channel value ranges from 1000 to 2000, representing the minimum and maximum stick deflection respectively, with 1500 corresponding to the neutral center position.

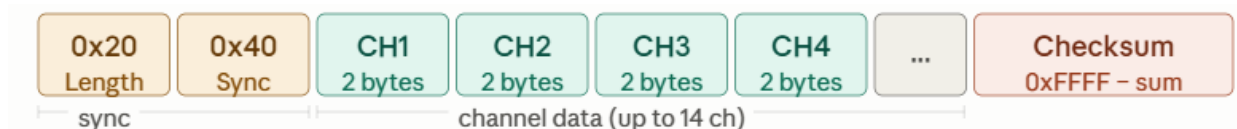


Figure 26 IBUS packet - 32 bytes every 140 Hz

5. Experimental Results

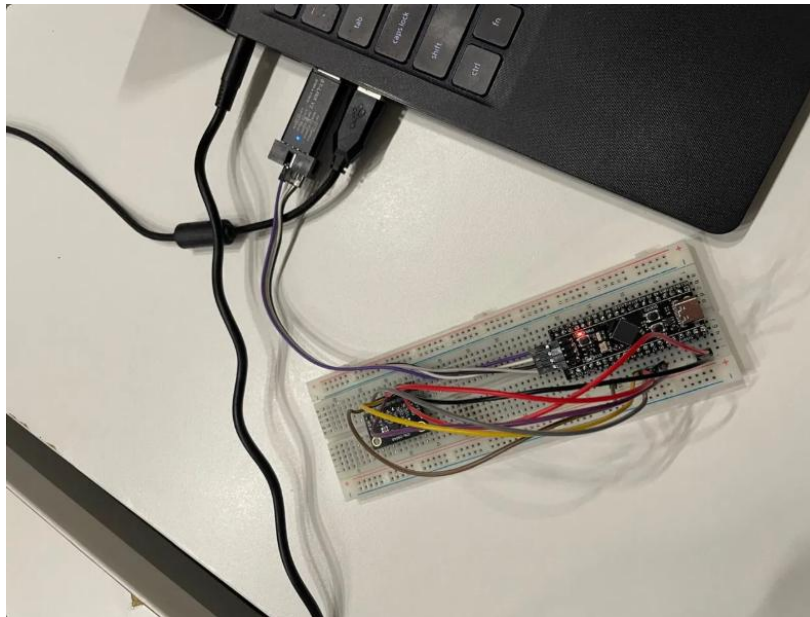


Figure 27 Development hardware setup: STM32F411CEUx (Black Pill) mounted on breadboard with ICM-20948 IMU

Table 9 Complete Pin Connection Table

STM32 Pin	STM32 Function	ICM-20948 Pin	ICM-20948 Function
PA4	SPI1_CS	CS	Chip select (active low)
PA5	SPI1_SCK	SCL	SPI Clock
PA6	SPI1_MISO	SDO	Slave Data Out
PA7	SPI1_MOSI	SDI	Slave Data in
3.3V	Power	VDD	Supply voltage
GND	Ground	GND	Ground

The STM32F411CEUx microcontroller serves as the central processing unit. The ICM-20948 9-axis IMU module is connected to the STM32 via the SPI1 peripheral, using four data lines: the chip select signal on PA4, the SPI clock on PA5, the MISO line on PA6 for data reception from the sensor, and the MOSI line on PA7 for transmitting commands and register addresses to the sensor. The sensor is powered from the 3.3V rail of the STM32 board.



Figure 28 My drone after soldering the ESCs to the frame

Figure 25 shows the quadcopter at an intermediate stage of assembly, after the four ESCs had been soldered to the Power Distribution Board (PDB) integrated into the lower plate of the frame. Each ESC was connected to the corresponding motor and to the PDB, which distributes power from the battery to all four motors through a common set of copper traces. This soldering step was critical, as a poor or cold joint at the PDB could interrupt the power supply to a motor and compromise the stability of the entire platform during flight.

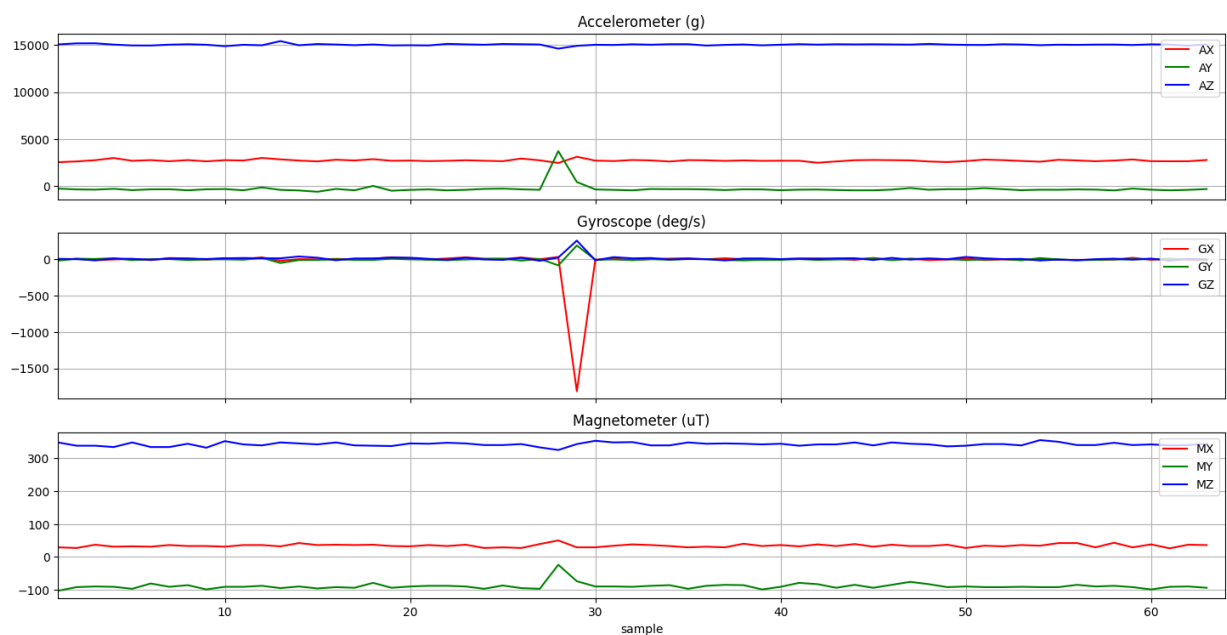


Figure 29 The raw output of all three ICM-20948 sensor axes

The graphs present the raw ADC output of all three sensors while the IMU is lying flat and stationary, and several issues are apparent: AX reads 2500 counts when it should read 0 — this is a DC offset bias that accumulates into angle error, AZ reads ~ 15000 instead of the theoretical 16384 for 1g — scale factor deviation. Both errors cause wrong roll and pitch if not corrected. On the gyroscope all axes are near zero which is good for a static sensor. The massive spike at sample ~ 28 reaching -1750 deg/s is a data anomaly. The magnetometer readings show large DC offsets on all three axes — $MX \approx +30 \mu\text{T}$, $MY \approx -100 \mu\text{T}$, $MZ \approx +330 \mu\text{T}$ — which are caused by hard-iron distortions from ferromagnetic materials in the vicinity of the sensor and would produce a systematic error in the yaw angle estimate.

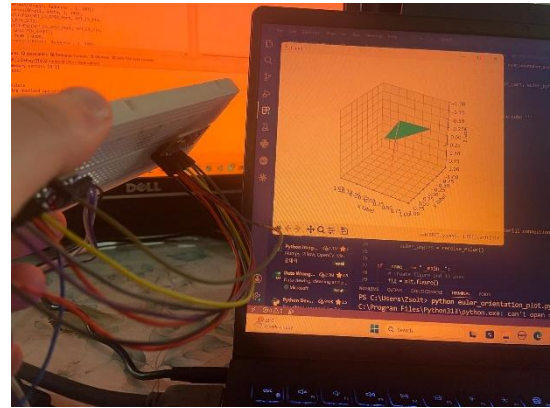
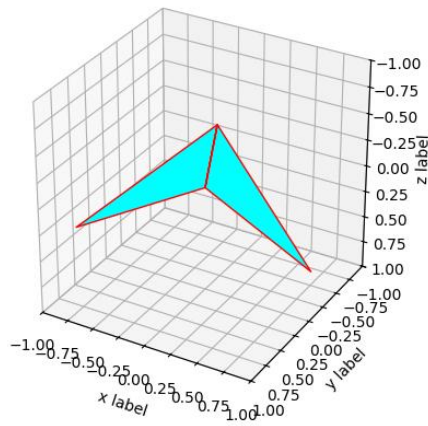


Figure 30 Real-time 3D visualization of the drone's orientation

This figure depicts a Python-based three-dimensional visualisation tool. The application establishes a serial link with the STM32 microcontroller through the USART2 peripheral, receiving the Euler angles—roll, pitch and yaw—computed by the onboard complementary filter at a rate of 100 Hz. Upon reception of each data frame, the software renders an updated representation of a geometrically simplified drone body as a three-dimensional polygon within a Matplotlib interactive axes environment, thereby providing a continuous and visually intuitive depiction of the estimated orientation. Throughout the development and integration phases, this visualisation tool functioned as the primary diagnostic instrument, enabling immediate and unambiguous verification of several critical aspects of the system: the correct alignment of the IMU measurement axes with respect to the body frame definition, the proper application of the coordinate frame remapping required to reconcile the differing axis orientations of the accelerometer and gyroscope subsystem relative to the AK09916 magnetometer die, the numerical stability and accuracy of the complementary filter output under both static and dynamic conditions, and the integrity of the UART data transmission pipeline between the STM32 and the host computer at the nominal update rate of 100 Hz.

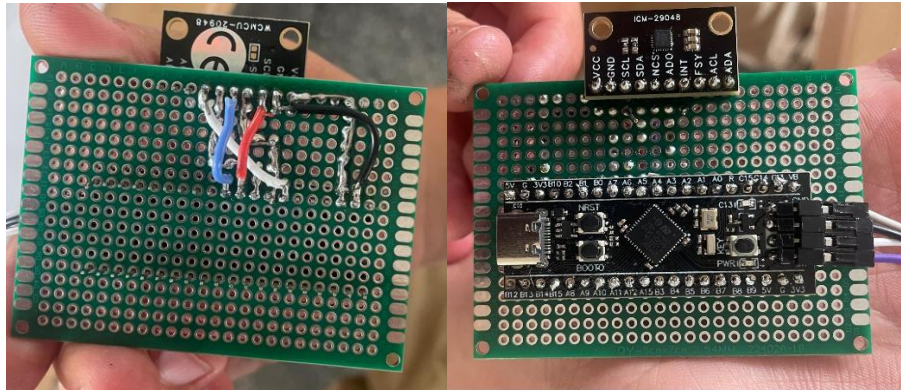


Figure 31 Soldered PCB

This figure shows the circuit after it had been transferred to a permanent soldered assembly in order to obtain a more mechanically robust, flight-ready hardware platform. Consolidating all flight-critical electronics onto a single compact board substantially reduces wiring complexity, improves mechanical reliability under vibration and facilitates integration into the drone airframe. The USB-C port of the Black Pill module remains accessible for firmware programming and serial debugging.

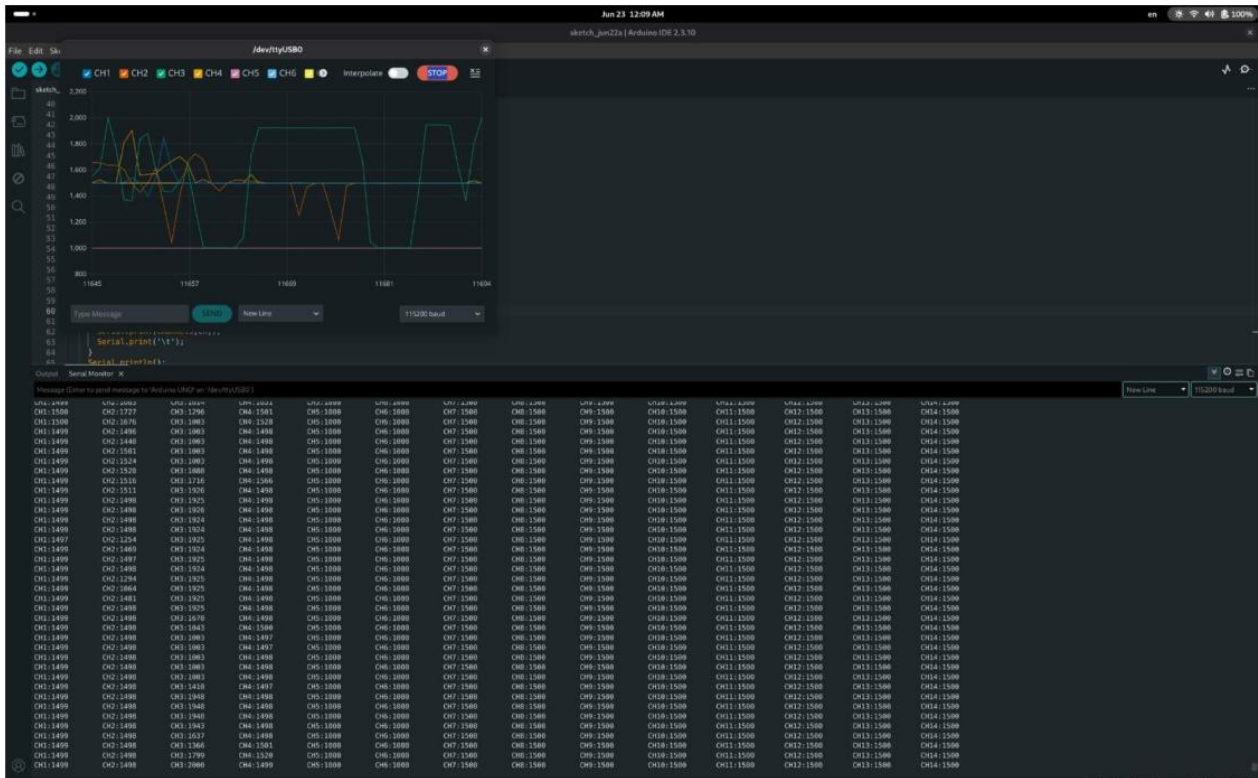


Figure 32 Integrating RC joystick

Before integrating the radio link into the flight controller, the FlySky receiver was tested in the Arduino IDE in order to verify that the transmitter signals were being received and decoded correctly. The microcontroller read the PWM signal of each channel and printed the pulse width, expressed in microseconds, to the serial port. As shown in the corresponding figure, the values were then visualised in two complementary ways: the Serial Monitor displayed the numerical output of all channels, whereas the Serial Plotter rendered the first six channels graphically in real time.

Expression	Type	Value
pwm_duty_percentage		Failed to evaluate expre
(*)= compare_value	uint16_t	1000
imu_sensor		Failed to evaluate expre
dumb_data		Failed to evaluate expre
sum_gyro_x		Failed to evaluate expre
sum_gyro_y		Failed to evaluate expre
sum_gyro_z		Failed to evaluate expre
> imu_norm_var	imu_norm	{...}
> euler_temp	euler_angles	{...}
(*)= ibus_channels.ch[0]	uint16_t	2000
(*)= ibus_channels.ch[1]	uint16_t	1923
(*)= ibus_channels.ch[2]	uint16_t	1999
(*)= ibus_channels.ch[3]	uint16_t	1998
(*)= ibus_channels.ch[4]	uint16_t	2000
(*)= ibus_channels.ch[5]	uint16_t	1000
ch1_roll		Failed to evaluate expre
ch3_throttle		Failed to evaluate expre
ch4_yaw		Failed to evaluate expre
(*)= motor1_command	float	0
(*)= motor2_command	float	0
(*)= timer_flag	volatile uint8_t	0 '\0'
(*)= overrun	volatile uint32_t	0
(*)= dma_callback_count	volatile uint32_t	5088
(*)= spi_dma_busy	volatile uint8_t	0 '\0'
(*)= ibus_index	volatile uint8_t	3 '\003'
(*)= ibus_buffer[0]	volatile uint8_t	32 ''
(*)= motor3_command	float	0
(*)= motor4_command	float	0
motor4_duty		Failed to evaluate expre
motor1_duty		Failed to evaluate expre
motor2_duty		Failed to evaluate expre
motor3_duty		Failed to evaluate expre
(*)= drone.start_flight	uint8_t	0 '\0'
(*)= drone.voltage_default	float	0

Figure 33 Live Expression

In order to verify the correct operation of the firmware directly on the microcontroller, the *Live Expressions* function of STM32CubeIDE was used. This feature allows a selected set of variables to be observed in real time during a debug session: the ST-LINK probe reads their values from the memory of the microcontroller through the SWD interface and updates them periodically, without interrupting the execution of the program. In this way, it is possible to confirm that the different software modules produce the expected values while the system is actually running.

The figure shows the state of the main variables during one of these tests. The fields `ibus_channels.ch[0]` to `ibus_channels.ch[5]` contain the values transmitted by the radio receiver through the iBus protocol; their range, situated approximately between 1000 and 2000, corresponds to the standard interval of the RC channels and demonstrates that the commands of the remote control are received and decoded correctly. The variable `dma_callback_count`, which had reached the value 5088, indicates that the reception based on DMA had already been executed several thousand times, and therefore that the serial communication with the receiver was continuous and stable. The counter `overrun`, equal to zero, confirms in addition that no reception error had occurred.

It should be noted that some expressions are marked as *"Failed to evaluate expression"*. This does not represent a malfunction, but it is a consequence of the fact that these variables are local to a specific function and are therefore out of scope at the moment of the reading, or that they were eliminated by the optimisation of the compiler. Only the global and static variables can be observed in a reliable manner with this method. For this reason, the variables that had to be monitored during the tests were declared as global, so that their evolution could be followed throughout the whole execution.

Each channel produced a pulse width in the expected range of approximately 1000 to 2000 μs , with a neutral position of around 1500 μs . When a stick on the transmitter was held in its central position, the corresponding channel remained close to 1500 μs ; moving the stick towards either extreme caused the value to rise towards 2000 μs or fall towards 1000 μs , which is clearly reflected in the fluctuations visible in the plotter. The unused channels stayed at their default value of 1500 μs . This confirmed that the receiver was functioning correctly and that the channel mapping was consistent, which was a necessary precondition before the signals could be used for motor control.



Figure 34 First test of the drone

Figure 33 shows the first physical test of the drone that was assembled for this project. This first test was carried out on the ground, before any real flight, and its purpose was mainly to verify that the assembly had been performed correctly. At this stage, the objective is to check that the motors receive the commands and rotate in the correct direction, that the connections are stable, and that the whole system is powered in a proper way. Such a preliminary test is an important step, because it allows possible errors in the wiring or in the configuration to be detected in a saafe manner, without the risk of a fall, before the drone is actually allowed to fly.

6. Conclusion

This project has demonstrated the successful design and implementation of a quadcopter UAV platform built upon the STM32F411CEU microcontroller, integrating a 9-axis ICM-20948 inertial measurement unit and a custom flight control firmware developed entirely from low-level embedded systems principles. Throughout the development process, core competencies were established in SPI communication, direct memory access, real-time attitude estimation using complementary filtering and quaternion mathematics, and closed-loop PID control. The theoretical foundations presented — spanning agricultural challenges, drone taxonomy, spectral imaging principles, and sensor fusion — provide the necessary context for understanding the significance of UAV technology in precision agriculture.

Nevertheless, the current implementation is subject to several notable limitations. First, the absence of a barometric altimeter means that altitude is not actively controlled, making stable hovering dependent solely on manual throttle input. Second, the platform lacks any GPS module, which prevents autonomous navigation, waypoint following, or position hold. Third, the complementary filter, while computationally efficient, is sensitive to prolonged linear accelerations and magnetic interference, which can introduce drift in the roll, pitch, and yaw estimates. Fourth, the flight endurance is constrained by the 3S 2200 mAh lithium polymer battery, which provides approximately 15–20 minutes of operation under typical load conditions. Finally, the current system does not implement any obstacle avoidance mechanism, limiting safe operation to open, controlled environments.

Looking ahead, several development directions are planned to expand the capabilities of the platform. In the near term, the integration of a barometric pressure sensor will enable altitude-hold functionality, while the addition of a GPS module will make it possible to implement a return-to-home feature — a critical safety mechanism that allows the drone to autonomously navigate back to its takeoff location in the event of signal loss or low battery. These two additions will represent the transition from a manually piloted vehicle to a semi-autonomous one.

In the longer term, the most significant planned extension is the integration of a Raspberry Pi single-board computer alongside a camera module. This addition will transform the platform from a flight control experiment into a functional precision agriculture tool. The Raspberry Pi will serve as a companion computer responsible for higher-level processing tasks, offloading computationally intensive algorithms from the STM32. Specifically, a field mapping algorithm will be developed to generate georeferenced aerial maps of agricultural land by stitching together sequential frames captured during autonomous flight paths. Building upon this mapping capability, a crop health analysis pipeline will be implemented, processing the captured imagery to compute vegetation indices and identify spatial patterns indicative of stress, disease, or nutrient deficiency. Together, these features will bring the system considerably closer to the class of precision agriculture drones described in the state-of-the-art review, and will provide a meaningful platform for continued research into low-cost, open-source agricultural UAVs.

Finally, it should be mentioned that a generative artificial intelligence tool was used during the preparation of this report as a support for language and phrasing. Its role was limited to the linguistic level, that is, to improving the clarity, the grammar and the academic formulation of the text, as well as to assisting with the correct wording of certain technical explanations.